CARRY-AWARE DIFFERENTIAL BOUNDS AND CERTIFIED CORRECTION OF LOCAL ZERO SUMS

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ABSTRACT. We study a two-digit coordinate model of maps on the Galois ring $\mathrm{GR}(4, m)$, with $q = 2^m$ and $N = q^2 = |\mathrm{GR}(4, m)|$, in which the ring addition carries from the low digit into the high digit. For maps $F(a, b) = (p(a),\ h(a) \oplus d(a)b)$ with p an APN permutation, d a nowhere-vanishing $\mathbb{F}_2$-affine map whose linear part has a one-dimensional kernel, and h arbitrary, we prove direction-wise differential counts for $B = J \circ F$, a key-averaged two-block bound of order $1/N$, and a collision-energy bound, all uniform in h. In contrast, a local integral property of the same two-block map depends on h; for the inverse map and every admissible d we give a linear certificate on h that excludes a family of local zero sums, and an explicit correction of Boolean degree two of every h that fails it; degree two is minimal for this local problem. For the averaged alternating two-word arrangement we prove a one-word contraction of order $q^{-(1-\frac{1}{2}\log_2 3)}$, uniform in h and in the admissible data, and deduce that a fixed number of steps brings the averaged two-word difference distribution to the scale $1/N$. For the inverse map with $d(a) = 1 + (a/\rho)^4 + (a/\rho)^2$, where m is odd and $\rho \neq 0$, the contraction improves to the order $q^{-1/2}$, which is optimal since no admissible data give a faster order; for other d our constant grows with the linearized degree of d. We then record what these statements do and do not imply for an independent-key composition: the averaged multi-word bound coexists with a candidate-existence distinguisher with data linear in the word size. No security claim for a concrete cipher is made.

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1. Introduction

1.1. **Two kinds of statements about the same map.** Let $R = \mathrm{GR}(4, m)$ be the Galois ring of characteristic 4 and residue field $\mathbb{F}_q$, $q = 2^m$, and put $N = q^2 = |R|$. Writing $x \in R$ in base 2 along a fixed basis gives two digits $x = (a, b) \in \mathbb{F}_q^2$; ring addition is then the digit-wise exclusive or, plus a carry $a \star u$ from the low into the high digit (Lemma 2.1). A polynomial map of R has, in these coordinates, the shape

$$F(a, b) = \big(p(a),\, h(a) \oplus d(a)\, b\big)$$

(Proposition 2.2); the swap $J(a, b) = (b, a)$ turns F into a map $B = J \circ F$ whose low digit depends on b. The present paper separates two kinds of statements about B.

Differential statements are uniform in h. Under an assumption on (p, d) only (Assumption 2.3), the differential uniformity of B with respect to ring subtraction is exactly q for every h (Theorem 3.4); the key-averaged differential probability of a two-block local map is at most $4/N$ when the kernel direction ρ of d has weight one, and at most $(6 + 6/q)/N$ in general, again for every h (Theorem 3.7); and a collision energy is bounded by an absolute constant (Theorem 3.9). The proofs are direction-wise counts that keep the carry visible; no change of basis is used.

Contraction is uniform in h. For the key-averaged alternating two-word arrangement, the one-word fluctuation operator of B has norm τ with $\tau^2 \le 5(3/4)^m - 3/q + 1/q^2$ for all PD data and all h (Theorem 6.2), so a fixed number of steps brings the difference distribution to the scale $1/N$ (Theorem 6.9). For one explicit family, namely $p(x) = x^{q-2}$, m odd, and $d = 1 \oplus K_0(x/\rho)$ with $K_0(X) = X^4 + X^2$ and any $\rho \neq 0$, one has $\tau^2 < 13.5/q$ for every h (Corollary 6.6); since one block of the operator has norm exactly $((q/2 - 1)/q^2)^{1/2}$ for all PD data (Proposition 6.3), the exponent $1/2$ is optimal for this family. For the inverse map with another admissible $d = \varepsilon \oplus K$, $K = \sum_{j=0}^{e} k_j X^{2^j}$, $k_e \neq 0$, the same argument gives only $\tau^2 \le (2^{e+1} + 5)/q + (q/2 - 1)/q^2$ (Proposition 6.5, Corollary 6.6), whose constant grows with e; the exponent $1/2$ uniformly over all PD data, or over all admissible d for the inverse map, is not proved (Remark 6.7).

Integral statements depend on h. The ring sum of the low digit of the same two-block map over a coset $a + 2R$ is a sum of a non-affine image (Section 4); whether it vanishes identically depends on h. For $p(x) = x^{q-2}$ and every admissible d we give an $\mathbb{F}_q$-linear functional ℓ_d on the space of all h whose non-vanishing excludes the local zero sum for every local key, and an explicit correction g_d of Boolean degree two, whose multiplicative complexity on its own is exactly $m - 1$, for every h with $\ell_d(h) = 0$; affine h

never suffice. The differential statements are untouched by this correction because they never depended on h.

The third part records the consequences, and the limits, for an independent-key composition with three mixing layers and local lengths $(2, 2, 1, 1)$: a multi-word averaged bound (Theorem 5.3) holds simultaneously with a candidate-existence distinguisher whose data is linear in N (Theorem 5.7). We use this to state precisely why a small averaged differential bound is not a PRP-security statement.

Table 1 lists the quantifiers of the main statements. "PD" means all (p, d) under Assumption 2.3.

TABLE 1. Main statements and their quantifiers.

statement	family	all h	ρ	exponent / constant / steps
Thm. 3.4	PD	yes	any	$\Delta(B) = q$, one step
Thm. 3.7	PD, odd $m \geq 5$	yes	$\mathrm{wt}(\rho) = 1$; any	$4/N$; $(6+6/q)/N$ (two blocks)
Thms. 4.5, 4.6	$p = x^{q-2}$, every admissible d, $m \geq 3$	yes*	any	certificate ℓ_d; correction g_d
Thm. 6.2	PD, $m \geq 3$	yes	any	$\tau \leq \sqrt{5}\, q^{-\gamma}$, $\gamma = 1 - \frac{1}{2}\log_2 3$
Prop. 6.3	PD, $m \geq 3$	yes	any	$\tau \geq ((q/2-1)/q^2)^{1/2}$: no exponent $> 1/2$
Cor. 6.6	$p = x^{q-2}$, $d = 1 \oplus K_0(x/\rho)$, odd m	yes	any $\rho \neq 0$	$\tau^2 < 13.5/q$: exponent $1/2$ (optimal)
Cor. 6.6 (general)	$p = x^{q-2}$, odd m, $\deg_2 K = e$	yes	any	$\tau^2 \leq (2^{e+1}+5)/q + (q/2-1)/q^2$
Thm. 6.9	PD, $m \geq 5$, A1	yes	any	$\|D_{14}\|_2 \leq 1\,600\,000/N$; zero word after 15 steps
Cor. 6.11	$p = x^{q-2}$, $d = 1 \oplus K_0(x/\rho)$, odd $m \geq 5$, A1	yes	any $\rho \neq 0$	$\|D_{13}\|_2 \leq 24\,405/N$; zero word after 14 steps
Thm. 5.3	PD, $q \geq 32$, $w \geq 2$, $(2, 2, 1, 1)$	yes	$\mathrm{wt}(\rho) = 1$; any	$(4/N)^w$; $((6+6/q)/N)^w$

*The certificate is a condition on h; the correction $h \mapsto h + g_d$ applies to every h that fails it. $\deg_2 K = e$ means $K = \sum_{j \leq e} k_j X^{2^j}$ with $k_e \neq 0$. All bounds concern key-averaged transitions.

1.2. **Relation to two earlier papers.** The papers [1] and [2] concern standard representatives of APN permutations and their Galois-ring lifts. The table lists the results of those papers that are used or recalled here, and how the present paper differs.

Paper	Result recalled	Present paper
[1]	Every quadratic APN permutation in odd dimension $m \geq 5$ is linearly equivalent to an optimal one ([1, Main Theorem]; existence).	Used only as background; no result here depends on it.
[2]	Explicit reduced representatives $P_{m,r}$, permutation and APN, with derivative $P' = 1+X^{2^{r-2}}+X^{2^{m-2}}$, whose linear part has kernel $\{0, \rho\}$, $\rho = 1$ ("derivative-optimal") and product formula $P_{m,r} + 1 = DE$ ([2, Thm. 3.1, Def. 2.1]); integer product form ([2, Prop. 5.1]); coefficient lifts are permutations ([2, Prop. 6.1]); $\delta_F \leq 2$ for unit directions $\bar{\alpha} \notin \{0, \rho\}$ ([2, Thm. 6.2(1)]).	The data (p, d, h) is arbitrary subject to Assumption 2.3; d need not be p', p need not be quadratic; the $k = 2$ direction bound is reproved (Proposition 3.3); the objects are key-averaged transitions and local sums, not representatives.

1.3. **Contributions in three groups.**

(I) *Differential structure common to all corrections h* (Section 3). Direction-wise counts for the classes $\mathcal{Z}, \mathcal{E}, \mathcal{G}$, the exact uniformity $\Delta(J \circ F) = q$, the two-block bound with the kernel direction made explicit, a uniform collision-energy bound, and (Section 6) the uniform one-word contraction R1 with the exact $\mathcal{Z}$ block, the h-free estimate of the $\mathcal{G}$ block, the resulting unconditional step count for the averaged alternating arrangement, and the exponent $1/2$ for the inverse map with $d = 1 \oplus K_0(x/\rho)$.

(II) *Constructive correction of local zero sums* (Section 4). For the inverse map and every non-zero two-to-one affine d: a linear certificate on h, an explicit correction of Boolean degree two, the exact pass fraction $1 - 1/q$ on every linear design space containing it, minimality of Boolean degree two for this local problem, and the count of cosets on which the local sum is non-zero.

(III) *Composition guarantees and what they do not prevent* (Section 5). The averaged multi-word bound for the $(2, 2, 1, 1)$ arrangement, the candidate-existence distinguisher and its conditioned false-acceptance bound, and the resources $T/D/M$ of the implemented enumeration.

Group (III) consists of applications of known principles (key averaging of Markov ciphers [3], the two-layer MDS count [5, 6], post-composition with a reset key, the zero sum of all ring elements [7], and the RP/RF comparison of [8]). It is included to fix the design meaning and the limits of (I) and (II), not as an independent novelty claim.

1.4. **What is not claimed.** We do not propose a cipher and do not claim that any construction in this paper is secure. In particular we do not claim: PRP security of any arrangement; a fixed-key differential bound; a key

schedule; a lower bound on the cost of any attack; that no attack below the codebook exists; minimality of AND counts of F, of S_t or of any arrangement (Proposition 4.9 concerns the correction alone); an advantage in cost over known Feistel or SPN constructions (for instance a three-round Feistel network with x^3 attains the same constant $4/N$ with fewer AND gates [4]). The contraction results of Section 6 concern key-averaged difference distributions of the alternating arrangement; they are not fixed-key statements and do not recommend a number of rounds. Optimality of the exponent $1/2$ refers to the order of decay in q for the family of Corollary 6.6, not to the constants, to a cipher, or to any other choice of d.

1.5. **Related work.** The analysis uses standard tools: branch-number (MDS) arguments [5, 6], Parseval's identity, Newton's identities for power sums, the Schur test, and the key-averaging view of Markov ciphers [3], under which the expected differential transition matrix is a Markov matrix whose subdominant spectrum governs convergence to uniformity (cf. [10]). What is specific here is the object and the explicit, family-uniform bound: for $B = J \circ F$ on $\mathrm{GR}(4, m)$ the carry-aware difference table restricted to the fluctuation subspace has norm τ with $\tau^2 \le 5(3/4)^m - 3/q + 1/q^2$; in the sources we read the subdominant eigenvalue is left implicit or assumed, not bounded in this way. The work on sum-free functions [9, 11] concerns sums of a function over k-dimensional affine subspaces; here the sums are of the inverse map over a non-affine image set, their non-vanishing is certified by a linear functional built from reciprocal power sums over the trace-one hyperplane, and a Boolean-degree-two correction makes the certificate non-zero for every h. We did not find these statements in the works cited; our search was limited, and the techniques themselves are not claimed as new.

2. Two-digit coordinates, carry, and the PD assumption

2.1. **Field, ring and coordinates.** Let $m \ge 3$, $q = 2^m$, and let $\mathbb{F}_q = \mathbb{F}_2[t]/(f)$ with f irreducible of degree m. We fix the polynomial basis $1, t, \dots, t^{m-1}$ once and for all and identify $\mathbb{F}_q$ with $\mathbb{F}_2^m$ through it. We write $\oplus$ for addition in $\mathbb{F}_q$, juxtaposition for the field product, and $a \star u$ for the coordinate-wise product in the fixed basis (bitwise AND). The operation $\star$ distributes over $\oplus$ and $u \star u = u$. For $x \in \mathbb{F}_q$, $\mathrm{wt}(x)$ is the number of non-zero coordinates.

Let $R = \mathrm{GR}(4, m) = \mathbb{Z}_4[t]/(\hat{f})$ with $\hat{f}$ a monic lift of f, and $N = |R| = q^2$. Every $x \in R$ is uniquely $x = \sum_i c_i t^i$ with $c_i = a_i + 2b_i \in \{0, 1, 2, 3\}$; with $a = (a_i)$, $b = (b_i) \in \mathbb{F}_q$ and $\sigma(a) = \sum a_i t^i$ we have $x = \sigma(a) + 2\sigma(b)$ and we write $x = (a, b)$. The reduction $x \mapsto a$ is the ring homomorphism $R \to \mathbb{F}_q$.

Lemma 2.1 (Carry)**.** $(a, b) + (u, v) = (a \oplus u,\ b \oplus v \oplus (a \star u))$ *and* $-(a, b) = (a,\ a \oplus b)$*. For* $y = (y_0, y_1)$*,* $z = (z_0, z_1)$ *and* $s = y_0 \oplus z_0$*,*

$$y - z = \big(s,\ y_1 \oplus z_1 \oplus (z_0 \star s)\big).$$

Proof. Coefficient-wise $a_i + u_i = (a_i \oplus u_i) + 2a_i u_i$ in $\mathbb{Z}_4$, and $2\sigma(x) + 2\sigma(y) = 2\sigma(x \oplus y)$ because $4 = 0$. This gives the sum. For the difference, $z + (s, w) = (y_0,\ z_1 \oplus w \oplus (z_0 \star s))$ by the sum formula, which equals y exactly when $w = y_1 \oplus z_1 \oplus (z_0 \star s)$. The negation is the case $y = 0$. □

In particular a difference with $s = 0$ has XOR high digit, while for $s \neq 0$ the high digit carries the correction $z_0 \star s$, which takes $2^{\mathrm{wt}(s)}$ values as z_0 varies.

2.2. **Polynomial maps in coordinates.**

Proposition 2.2 (Coordinate form and realizability)**.** *Let $\mathsf{F} \in R[X]$ and let $\bar{\mathsf{F}} \in \mathbb{F}_q[X]$ be its reduction. Then*

$$\mathsf{F}(a, b) = \big(\bar{\mathsf{F}}(a),\ h(a) \oplus d(a)\, b\big), \qquad d = \bar{\mathsf{F}}' \text{ (formal derivative)},$$

where $h(a)$ is the high digit of $\mathsf{F}(\sigma(a))$. Conversely, for arbitrary functions $p, d, h : \mathbb{F}_q \to \mathbb{F}_q$ there is $\mathsf{F} \in R[X]$ with $\mathsf{F}(a, b) = (p(a), h(a) \oplus d(a)b)$.

Proof. For $y, z \in R$, $(y + 2z)^n = y^n + 2nzy^{n-1} + 4(\cdots)$, hence $\mathsf{F}(y + 2z) = \mathsf{F}(y) + 2z\mathsf{F}'(y)$. With $y = \sigma(a)$, $z = \sigma(b)$, the term $2\sigma(b)\mathsf{F}'(\sigma(a))$ equals $2\sigma(\bar{\mathsf{F}}'(a)b)$, and adding an element of $2R$ only XORs the high digit. For the converse let $p_{\mathrm{rep}}, d_{\mathrm{rep}}$ be the interpolating polynomials of degree $< q$ and put $G = p_{\mathrm{rep}} + (X^q - X)(d_{\mathrm{rep}} + p'_{\mathrm{rep}})$. On $\mathbb{F}_q$, $G = p$, and $G' = p'_{\mathrm{rep}} + (qX^{q-1} - 1)(d_{\mathrm{rep}} + p'_{\mathrm{rep}}) + (X^q - X)(\cdots)'$ equals $p'_{\mathrm{rep}} + d_{\mathrm{rep}} + p'_{\mathrm{rep}} = d$ on $\mathbb{F}_q$ (characteristic 2, $q \equiv 0$). Let F_0 be any coefficient lift of G, with high digit h_0. If $\hat{H}$ lifts an interpolant of $h \oplus h_0$, then $2\hat{H}(\sigma(a) + 2\sigma(b)) = 2\sigma(h(a) \oplus h_0(a))$, so $\mathsf{F} = \mathsf{F}_0 + 2\hat{H}$ has high digit $h(a) \oplus d(a)b$. □

Thus the quantifier "over all coefficient lifts" is the same as "over all functions h", and d is not tied to p. Let $J(a, b) = (b, a)$; J is a bijection of R which is not additive. For p, d, h as above put

$$F = F_{p,d,h} : (a, b) \mapsto (p(a),\ h(a) \oplus d(a)b),$$
$$B = J \circ F : (a, b) \mapsto (h(a) \oplus d(a)b,\ p(a)).$$

2.3. **The PD assumption.**

For $u \in \mathbb{F}_q$ write $D_u p(a) = p(a \oplus u) \oplus p(a)$. A map p is APN if for every $u \neq 0$ and every c the equation $D_u p(x) = c$ has either no solution or exactly the two solutions $\{x, x \oplus u\}$.

Assumption 2.3 (PD)**.**
- (P1) $p : \mathbb{F}_q \to \mathbb{F}_q$ is an APN permutation (not necessarily quadratic);
- (P2) $d = \varepsilon \oplus K$ with K $\mathbb{F}_2$-linear, $\ker K = \{0, \rho\}$ for some $\rho \neq 0$, and $\varepsilon \notin \operatorname{Im} K$;
- (P3) $h : \mathbb{F}_q \to \mathbb{F}_q$ is arbitrary.

We write $\omega = 2^{\mathrm{wt}(\rho)}$, where the weight is taken in the fixed basis, and $S_\rho = \{x \star \rho : x \in \mathbb{F}_q\}$, so $|S_\rho| = \omega$.

Lemma 2.4. *Under (P2): $d(a) \neq 0$ for all a; $d(a \oplus u) \oplus d(a) = K(u)$; $d(a \oplus \rho) = d(a)$; every non-empty fibre of d is a pair $\{a, a \oplus \rho\}$; $\operatorname{Im} d = \varepsilon \oplus \operatorname{Im} K$ is an affine hyperplane of $\mathbb{F}_q$ not containing 0, so $|\operatorname{Im} d| = q/2$.*

Proof. $d(a) = 0$ would mean $\varepsilon = K(a) \in \operatorname{Im} K$. The other statements follow from linearity of K and $\dim \ker K = 1$. □

Remark 2.5. The basis is fixed. Replacing p by $p(\rho x)$ would normalize $\rho = 1$, but this changes the map because J and the carry are defined in the fixed basis; we never do this. For the inverse $p(x) = x^{q-2}$ with m odd and $d_0(a) = (a^2+a+1)^2$ one has $\rho = 1$, $\omega = 2$ (Section 4). The inverse map is an APN permutation exactly for odd m; when an APN permutation is needed for all statements at once we therefore take $p(x) = x^{q-2}$ with $m \geq 5$ odd. The counting lemmas of Section 3 and Theorem 6.2 use only (P1)–(P3) and do not use the parity of m.

2.4. **Difference distribution and direction classes.** All differences are ring differences. For a finite abelian group X (below $X = R$ or $X = R^w$) and a map $G : X \to X$ put

$$\delta_G(\alpha, \beta) = \#\{x \in X : G(x + \alpha) - G(x) = \beta\},$$
$$T_G = \mathrm{DP}_G = \delta_G/|X|, \qquad \Delta(G) = \max_{\alpha \neq 0, \beta} \delta_G(\alpha, \beta).$$

The normalizing factor is N for one word and N^w for w words, so that every row of T_G sums to 1. For a word-wise map $G = (G_1, \ldots, G_w)$ one has $\delta_G = \prod_i \delta_{G_i}$ and $|R^w| = N^w$, hence $T_G(\alpha, \beta) = \prod_i T_{G_i}(\alpha_i, \beta_i)$; multi-word tables enter only in this product form (Proposition 2.6, Lemmas 5.1, 5.2, Section 6). If G is a permutation, T_G is doubly stochastic, $T_G(0, \cdot) = e_0$ and $T_G(\alpha, 0) = 0$ for $\alpha \neq 0$. A non-zero difference $\alpha = (u, v)$ belongs to the class

$$\mathcal{Z} : \ u = 0 \ (q-1 \text{ elements}), \qquad \mathcal{E} : \ u = \rho \ (q), \qquad \mathcal{G} : \ u \notin \{0, \rho\} \ (q(q-2)).$$

2.5. **Keys, local blocks and averaged probabilities.** Keys are added in R. For $t \geq 1$ the local block with t key words is

$$S_t(x; k_1, \ldots, k_t) = F\big(B(\cdots B(B(x) + k_1) \cdots) + k_t\big),$$

so S_t uses $t + 1$ evaluations of F and t of J. In the averaging model the keys, and a whitening key k_0 in front, are independent and uniform.

Proposition 2.6 (Key averaging; standard)**.** *For $\alpha, \gamma \in R$,*

$$\mathbb{E}_{k_0, \ldots, k_t} \mathrm{DP}_{x \mapsto S_t(x + k_0; k)}(\alpha, \gamma) = (T_B^t T_F)(\alpha, \gamma).$$

Proof. This is the Markov-cipher principle [3]: for fixed inputs $x, x + \alpha$, the value $x + k_0$ is uniform and the difference is α; after each B the addition of an independent uniform key makes the next input uniform and independent of the difference already produced, so the difference evolves by the transition matrix T_B, and finally by T_F. □

We write $\mathrm{EDP}_t = \max_{\alpha \neq 0, \gamma} (T_B^t T_F)(\alpha, \gamma)$.

2.6. **The** $(2,2,1,1)$ **arrangement.** Let $w \geq 2$, let M_1, M_2, M_3 be R-linear automorphisms of R^w, and let local layers act word-wise. The arrangement studied in Section 5 is

$$E(x) = S_1^{(4)}\Big(M_3\, S_1^{(3)}\big(M_2\, S_2^{(2)}(M_1\, S_2^{(1)}(x + k_0) + L_1) + L_2\big) + L_3\Big),$$

with state names $X_1 = S_2^{(1)}(x + k_0)$, $V_1 = M_1X_1 + L_1$, $X_2 = S_2^{(2)}(V_1)$, $V_2 = M_2X_2 + L_2$, $X_3 = S_1^{(3)}(V_2)$, $V_3 = M_3X_3 + L_3$, $Y = S_1^{(4)}(V_3)$. It has three mixing layers and two reset keys L_2, L_3 in front of the one-key layers. The key words are $k_0, L_1, L_2, L_3, K_3, K_4$ (w words each) and K_1, K_2 ($2w$ words each), $10w$ words in total, all independent and uniform in the averaging model. These $10w$ words are the randomness of the model, not a security level; no key schedule is specified. The arrangement and the number of mixing layers are fixed throughout; we do not add layers. We assume that every entry and every minor of M_1 is a unit (MDS) and that M_2, M_3 are invertible.

3. Differential structure common to all corrections

Throughout this section Assumption 2.3 holds, $F = F_{p,d,h}$ and $B = J \circ F$. No statement depends on h.

3.1. **Difference formulas.**

Lemma 3.1. *For $x = (a, b)$ and $\alpha = (u, v)$,*

$$F(x+\alpha) - F(x) = \Big(D_u p(a),\\ h(a \oplus u) \oplus h(a) \oplus K(u)b \oplus d(a \oplus u)(v \oplus a \star u) \oplus \big(p(a) \star D_u p(a)\big)\Big).$$

Proof. By Lemma 2.1, $x + \alpha = (a \oplus u,\ b \oplus v \oplus a \star u)$. Apply the difference formula of Lemma 2.1 with $z_0 = p(a)$, $s = D_u p(a)$, and use $d(a \oplus u)b \oplus d(a)b = K(u)b$. □

Lemma 3.2. *For $x = (a, b)$ and $\alpha = (u, v)$ put $Q = h(a) \oplus d(a)b$. Then $B(x + \alpha) - B(x) = (s, t)$ with*

$$s = h(a \oplus u) \oplus h(a) \oplus K(u)\, b \oplus d(a \oplus u)(v \oplus a \star u), \qquad t = D_u p(a) \oplus (Q \star s).$$

Proof. $B(x) = (Q, p(a))$ and $B(x+\alpha) = (Q', p(a \oplus u))$ with $Q' = h(a \oplus u) \oplus d(a \oplus u)(b \oplus v \oplus a \star u)$. Lemma 2.1 with $z_0 = Q$ and $s = Q' \oplus Q$ gives the claim. □

The output class of a difference of B is read off its low digit s: $s = 0$, $s = \rho$, or $s \notin \{0, \rho\}$. Since B is a permutation (by (P1) and $d \neq 0$), $\alpha \neq 0$ gives $(s, t) \neq 0$.

Proposition 3.3 (Rows of T_F)**.** *Let $\alpha = (u, v) \neq 0$.*

(1) *If $\alpha \in \mathcal{G}$ then $\delta_F(\alpha, \beta) \leq 2$ for all β.*

(2) *If* $\alpha \in \mathcal{Z}$ *then* $F(x+\alpha)-F(x) = (0, d(a)v)$*; hence* $\delta_F(\alpha,(0,w)) = 2q$ *if* $w/v \in \operatorname{Im} d$ *and* $\delta_F(\alpha,\beta) = 0$ *otherwise.*
(3) *If* $\alpha \in \mathcal{E}$ *then the high digit of* $F(x+\alpha)-F(x)$ *does not depend on* b *and, for fixed* a*, is a bijective function of* v*; in particular* $\delta_F(\alpha,\beta) \le 2q$*.*

Proof. (1) $K(u) \neq 0$, so for each a at most one b gives a prescribed high digit, and $D_u p(a) = \bar{\beta}$ has at most two solutions by (P1). (2) Put $u = 0$ in Lemma 3.1; d is two-to-one onto $\operatorname{Im} d$. (3) $K(\rho) = 0$ removes b, and the coefficient of v is $d(a \oplus \rho) = d(a) \neq 0$; at most two a solve $D_\rho p(a) = \bar{\beta}$, each with q values of b. □

3.2. **Exact differential uniformity.**

Theorem 3.4. $\Delta(B) = q$*; that is,* $\max_{\alpha\neq 0,\beta} T_B(\alpha,\beta) = 1/q$*. More precisely, for every* β*:* $\delta_B(\alpha,\beta) \le q$ *if* $\alpha \in \mathcal{G}$*;* $\delta_B(\alpha,\beta) \le 2q/2^{\mathrm{wt}(s)} \le q$ *if* $\alpha \in \mathcal{Z}$ *and* $\beta = (s,t)$*; and* $\delta_B(\alpha,\beta) \le q$ *if* $\alpha \in \mathcal{E}$*. The value* q *is attained.*

Proof. Use Lemma 3.2 with $\beta = (s,t)$.

$\mathcal{G}$. $K(u) \neq 0$, so for fixed a the value s determines b. Hence at most one b per a, at most q in total.

$\mathcal{Z}$. Here $s = d(a)v \neq 0$, so at most two a (fibres of d). For fixed a, Q is a bijective function of b and $t = Q \star s$; the map $Q \mapsto Q \star s$ takes $2^{\mathrm{wt}(s)}$ values, each $q/2^{\mathrm{wt}(s)}$ times. Total $\le 2q/2^{\mathrm{wt}(s)} \le q$ as $s \neq 0$.

$\mathcal{E}$. $K(\rho) = 0$, so $s = s(a)$ does not depend on b, and from $d(a \oplus \rho) = d(a)$ and $(a \oplus \rho) \star \rho = (a \star \rho) \oplus \rho$,

$$s(a \oplus \rho) = s(a) \oplus d(a)\rho \neq s(a). \tag{1}$$

Let $S_s = \{z \star s : z \in \mathbb{F}_q\}$. The condition is $s(a) = s$ and $D_\rho p(a) \in t \oplus S_s$. By (P1) every non-empty fibre of $D_\rho p$ is a pair $\{a, a \oplus \rho\}$, so the admissible a lie in at most $|S_s|$ pairs; by (1) each pair contains at most one a with $s(a) = s$; for that a exactly $q/|S_s|$ values of b give the required $Q \star s$. Total $\le q$.

Attainment. Let η be a basis vector, $d_* \in \operatorname{Im} d$, $v = \eta/d_*$, $\alpha = (0,v)$, $\beta = (\eta, 0)$. Then $s = \eta$ exactly for the two a with $d(a) = d_*$, and for each such a exactly $q/2$ values of b give $Q \star \eta = 0$. □

The proof uses neither $\mathrm{wt}(\rho)$ nor the permutation property of p in the cases $\mathcal{Z}, \mathcal{E}$.

3.3. **Class transitions.**

Proposition 3.5. *For* β *in the indicated class, the following column sums (over all rows* α *of the indicated class) and entry bounds hold for* T_B*:*

rows $\to$ *target*	*quantity*	*bound*	*ingredient*
$\mathcal{Z} \to \mathcal{Z}$	*column sum*	0	$s = d(a)v \neq 0$
$\mathcal{Z} \to \mathcal{E}$	*column sum*	$1/\omega$	*fibres of* d, Q *uniform in* b
$\mathcal{E} \to \mathcal{Z}$	*column sum*	$2/q$	$v \mapsto s_v(a)$ *bijective, APN in direction* ρ
$\mathcal{E} \to \mathcal{E}$	*column sum*	$2/q$	*as above,* $D_\rho p(a) \in t \oplus S_\rho$
$\mathcal{G} \to \mathcal{Z}$	*entry*	$2/N$	b *unique per* a, *APN*
$\mathcal{G} \to \mathcal{E}$	*entry*	$2\omega/N$	b *unique per* a, $D_u p(a) \in t \oplus S_\rho$

For the final layer F*: if* γ *has low digit* 0*, only rows in* $\mathcal{Z}$ *contribute and their column sum is at most* 1*; if* γ *has non-zero low digit, rows in* $\mathcal{Z}$ *contribute* 0*, rows in* $\mathcal{E}$ *have column sum at most* $2/q$*, and rows in* $\mathcal{G}$ *have entries at most* $2/N$*.*

Proof. All counts use Lemma 3.2 and divide by $N = q^2$. $\mathcal{Z} \to \mathcal{Z}$: $s = d(a)v \neq 0$. $\mathcal{Z} \to \mathcal{E}$, target (ρ, t): for each v with $\rho/v \in \operatorname{Im} d$ (there are $q/2$ such v) exactly two a have $s = \rho$; then $t = Q \star \rho$ and Q is uniform in b, so q/ω values of b (if $t \in S_\rho$). The column sum is at most $(q/2) \cdot 2 \cdot (q/\omega)/N = 1/\omega$. $\mathcal{E} \to \mathcal{Z}$, target $(0, t)$: for fixed a, $v \mapsto s_v(a) = h(a \oplus \rho) \oplus h(a) \oplus d(a)(v \oplus a \star \rho)$ is bijective, so one v per a gives $s = 0$; then $t = D_\rho p(a)$ has at most two solutions a, each with q values of b: column sum $\leq 2q/N$. $\mathcal{E} \to \mathcal{E}$, target (ρ, t): one v per a; $D_\rho p(a) \in t \oplus S_\rho$ for at most 2ω values of a, each with q/ω values of b: column sum $\leq 2q/N$. $\mathcal{G} \to \mathcal{Z}$: $s = 0$ fixes b for each a, and $t = D_u p(a)$ has at most two solutions. $\mathcal{G} \to \mathcal{E}$: $s = \rho$ fixes b, and $D_u p(a) \in t \oplus S_\rho$ has at most 2ω solutions. Final layer: by Lemma 3.1 the low digit of an F-difference is $D_u p(a)$, which is non-zero for $u \neq 0$ because p is a permutation; T_F is doubly stochastic, so any partial column sum is at most 1. For rows in $\mathcal{E}$ apply Proposition 3.3(3): at most two a, one v per a, q values of b. For rows in $\mathcal{G}$ apply Proposition 3.3(1). □

3.4. **A sup-norm recursion and the two-block bound.** Fix $\alpha \neq 0$ and let $\pi_t = e_\alpha T_B^t$ be the distribution of the difference after t key-separated applications of B (Proposition 2.6). Put $\sigma_t^{\mathcal{Z}} = \max_{\beta \in \mathcal{Z}} \pi_t(\beta)$ and $\sigma_t^{\mathcal{E}} = \max_{\beta \in \mathcal{E}} \pi_t(\beta)$.

Theorem 3.6 (Sup-norm recursion)**.** *For every* $t \geq 0$*,*

$$\sigma_{t+1}^{\mathcal{Z}} \leq \frac{2}{N} + \frac{2}{q}\,\sigma_t^{\mathcal{E}}, \qquad \sigma_{t+1}^{\mathcal{E}} \leq \frac{2\omega}{N} + \frac{1}{\omega}\,\sigma_t^{\mathcal{Z}} + \frac{2}{q}\,\sigma_t^{\mathcal{E}},$$

$$\max_\gamma (\pi_t T_F)(\gamma) \leq \max\Big(\sigma_t^{\mathcal{Z}},\ \frac{2}{N} + \frac{2}{q}\,\sigma_t^{\mathcal{E}}\Big).$$

Proof. $\pi_{t+1}(\beta) = \sum_{\alpha'} \pi_t(\alpha') T_B(\alpha', \beta)$ with $\pi_t(0) = 0$. Split α' by class. The rows in $\mathcal{G}$ contribute at most (largest entry)$\cdot \pi_t(\mathcal{G}) \leq$ largest entry; the rows in $\mathcal{Z}$ (resp. $\mathcal{E}$) contribute at most $\sigma_t^{\mathcal{Z}}$ (resp. $\sigma_t^{\mathcal{E}}$) times the column sum over that class. Insert Proposition 3.5. The last inequality is the same argument with the final-layer part of Proposition 3.5. □

Theorem 3.7 (Two-block bound, kernel direction explicit)**.** *Let $m \geq 5$ be odd. For every h and every $\alpha \neq 0$, γ:*

(1) *if* $\mathrm{wt}(\rho) = 1$ *(so $\omega = 2$), then* $(T_B^2 T_F)(\alpha, \gamma) \leq 4/N$*;*
(2) *for general ρ,* $(T_B^2 T_F)(\alpha, \gamma) \leq (6 + 6/q)/N \leq 99/(16N)$.

By Proposition 2.6, EDP_2 *of the local block S_2 obeys the same bounds.*

Proof. Put $x = 1/q$, so $x^2 = 1/N$. By Theorem 3.4, $\pi_1(\beta) = T_B(\alpha, \beta) \leq x$ for all β, hence $\sigma_1^{\mathcal{Z}}, \sigma_1^{\mathcal{E}} \leq x$. Theorem 3.6 uses of π_t only that it is a probability vector and the two maxima, so it may be started at $t = 1$: $\sigma_2^{\mathcal{Z}} \leq 2x^2 + 2x \cdot x = 4x^2$ and $\sigma_2^{\mathcal{E}} \leq 2\omega x^2 + x/\omega + 2x^2$. Therefore $(T_B^2 T_F)(\alpha, \gamma) \leq \max(4x^2,\ 2x^2 + 2x\sigma_2^{\mathcal{E}})$ and

$$N\big(2x^2 + 2x\sigma_2^{\mathcal{E}}\big) = 2 + \frac{2}{\omega} + \frac{4(\omega + 1)}{q}.$$

For $\omega = 2$ this is $3 + 12/q \leq 4$ since $q \geq 12$. For general $2 \leq \omega \leq q$, $(6+6/q) - \big(2+2/\omega+4(\omega+1)/q\big) = (4\omega-2)(q-\omega)/(q\omega) \geq 0$, and $4 \leq 6+6/q$. Finally $6 + 6/q \leq 99/16$ for $q \geq 32$. □

Remark 3.8. The constant 4 in (1) uses $\omega = 2$, i.e. weight one of ρ in the fixed basis; it is not obtained for general ρ by a change of basis (Section 2). All tables used in Sections 4–5 have $\rho = 1$.

3.5. **A uniform collision-energy bound.** Define the collision energy of B as $\mathcal{E}_B = \sum_{\alpha \in R} \sum_{\beta} T_B(\alpha, \beta)^2$ (the row $\alpha = 0$ included).

Theorem 3.9 (Uniform energy)**.** *Assume (P2), (P3) and that p is APN (the permutation property is not used). Then for every m, every h and every ρ,*

$$\mathcal{E}_B \leq 6 + 4(3/4)^m, \qquad \textit{and} \qquad \mathcal{E}_B \leq 1779/256 < 7 \quad \textit{if } m \geq 5.$$

The proof occupies the rest of this subsection. Write $\mathcal{E}_{\mathcal{Z}}, \mathcal{E}_{\mathcal{E}}, \mathcal{E}_{\mathcal{G}}$ for the contributions of the rows in each class; the zero row contributes 1.

Lemma 3.10 ($\mathcal{Z}$ rows, exact)**.** $\mathcal{E}_{\mathcal{Z}} = 2(3/4)^m - 2/q$.

Proof. For $u = 0$, Lemma 3.2 gives $s = d(a)v$ and $t = Q \star s$. Hence $\delta_B((0, v), (s, t)) = 2q/2^{\mathrm{wt}(s)}$ if $s \in v \operatorname{Im} d$ and $t \subseteq s$ (bitwise), and 0 otherwise. The row energy is $(4/q^2) \sum_{s \in v\operatorname{Im} d} 2^{-\mathrm{wt}(s)}$. Each $s \neq 0$ lies in $v\operatorname{Im} d$ for exactly $|\operatorname{Im} d| = q/2$ values of $v \neq 0$. Thus $\mathcal{E}_{\mathcal{Z}} = (4/q^2)(q/2) \sum_{s \neq 0} 2^{-\mathrm{wt}(s)} = (2/q)((3/2)^m - 1)$. □

Lemma 3.11 ($\mathcal{E}$ rows)**.** $\mathcal{E}_{\mathcal{E}} \leq 1$.

Proof. By the $\mathcal{E}$ case of Theorem 3.4, which uses only APN and (P2), every entry of an $\mathcal{E}$ row is at most $1/q$; rows sum to 1, so each row energy is at most $1/q$, and there are q rows. □

For the $\mathcal{G}$ rows fix $u \notin \{0, \rho\}$ and put

$$\lambda(a) = d(a)K(u)^{-1}, \qquad w(a) = K(u)\lambda(a)\lambda(a \oplus u),$$
$$H_u(a) = \lambda(a \oplus u)h(a) \oplus \lambda(a)h(a \oplus u).$$

Lemma 3.12. *For $\alpha = (u, v)$ and each (a, s) there is exactly one b with low difference s, and then*

$$\delta_B((u,v),(s,t)) = \#\{a : t_s(a) = t\}, \qquad t_s(a) = D_u p(a) \oplus \big(s \star z_s(a)\big),$$
$$z_s(a) = H_u(a) \oplus \lambda(a)s \oplus w(a)(v \oplus a \star u).$$

Moreover: (S1) $\lambda(a \oplus \rho) = \lambda(a)$, $\lambda(a \oplus u) = \lambda(a) \oplus 1$, $w(a \oplus \rho) = w(a \oplus u) = w(a) \neq 0$; (S2) $w(a') = w(a)$ *iff* $a' \in a \oplus \{0, \rho, u, u \oplus \rho\}$; (S3) $z_s(a) \oplus z_s(a \oplus u) = s \oplus w(a)u$; $z_s(a) \oplus z_s(a \oplus \rho) = Z_\rho(a)$ *and* $z_s(a) \oplus z_s(a \oplus \rho \oplus u) = s \oplus Z_{\rho u}(a)$, *where $Z_\rho(a), Z_{\rho u}(a)$ depend neither on s nor on v.*

Proof. Solve s in Lemma 3.2 for b (possible as $K(u) \neq 0$) and substitute in Q; use $\lambda(a \oplus u) = (d(a) \oplus K(u))K(u)^{-1} = \lambda(a) \oplus 1$, so $h(a)(1 \oplus \lambda(a)) = \lambda(a \oplus u)h(a)$ and $\lambda(a)d(a \oplus u) = w(a)$. (S1) follows from Lemma 2.4; $w(a) \neq 0$ because d has no zeros. (S2): $w = K(u)g(\lambda)$ with $g(x) = x^2 + x$, which is two-to-one with $g(x) = g(x \oplus 1)$; so $w(a') = w(a)$ iff $d(a') \in \{d(a), d(a \oplus u)\}$ iff $a' \oplus a \in \ker K$ or $a' \oplus a \oplus u \in \ker K$. (S3): $H_u(a \oplus u) = H_u(a)$ and $(a \star u) \oplus ((a \oplus u) \star u) = u$; in the other two differences λ and w take the same value at both points, so the $\lambda(a)s$ terms and the $w(a)v$ terms cancel (resp. leave exactly s). □

Lemma 3.13 (Energy identity)**.** *With $A_u = \sum_a 2^{\mathrm{wt}(w(a)u)}$, $\mathrm{Merge}_u = \#\{(s, a, a') : a' \in \{a \oplus \rho, a \oplus \rho \oplus u\},\ t_s(a) = t_s(a')\}$ (independent of v) and $\mathrm{Cross}_u = \sum 3^{m-\mathrm{wt}(r)}$ over ordered pairs (a, a') with $w(a) \neq w(a')$, $r = D_u p(a) \oplus D_u p(a')$,*

$$\sum_v \sum_{s,t} \delta_B((u,v),(s,t))^2 = q^3 + q\,A_u + \mathrm{Cross}_u + q\,\mathrm{Merge}_u.$$

Proof. For each s the row has exactly q points with low difference s (one per a), so $\sum_t \delta_B^2 = \#\{(a, a') : t_s(a) = t_s(a')\}$. Split the ordered pairs: (i) $a' = a$ gives q^3 after summing over v, s, a. (ii) $a' = a \oplus u$: $D_u p(a \oplus u) = D_u p(a)$ and the condition is $s \star (s \oplus w(a)u) = 0$, i.e. $s \subseteq w(a)u$, which holds for $2^{\mathrm{wt}(w(a)u)}$ values of s and every v. (iii) $a' \in \{a \oplus \rho, a \oplus \rho \oplus u\}$: by (S3) the condition does not depend on v. (iv) otherwise $w(a) \neq w(a')$ by (S2), and $z_s(a) \oplus z_s(a') = c \oplus (w(a) \oplus w(a'))v$ with c independent of v; the condition $r = s \star (c \oplus (w \oplus w')v)$ has $2^{m-\mathrm{wt}(s)}$ solutions v if $r \subseteq s$ and none otherwise, and $\sum_{s \supseteq r} 2^{m-\mathrm{wt}(s)} = 3^{m-\mathrm{wt}(r)}$. □

Lemma 3.14. $\mathrm{Merge}_u \leq q^2$, $A_u \leq 4(3^m - 1)$, $\mathrm{Cross}_u \leq 2q(q^2 - 3^m)$.

Proof. Merge. Let $r_a = D_u p(a) \oplus D_u p(a \oplus \rho)$. Since $a \oplus \rho \notin \{a, a \oplus u\}$, APN gives $r_a \neq 0$, and $D_u p(a \oplus \rho \oplus u) = D_u p(a \oplus \rho)$ gives the same r_a for the second partner. For $a' = a \oplus \rho$ the condition is $r_a = s \star Z_\rho(a)$, with $2^{m-\mathrm{wt}(Z_\rho)} \leq q/2$ solutions (as $\mathrm{wt}(Z_\rho) \geq \mathrm{wt}(r_a) \geq 1$ whenever solutions exist). For $a' = a \oplus \rho \oplus u$ the condition is $r_a = s \star \overline{Z_{\rho u}}$ (bitwise complement), with $2^{\mathrm{wt}(Z_{\rho u})} \leq q/2$ solutions. Hence $\mathrm{Merge}_u \leq q \cdot 2 \cdot q/2$. *$A_u$.* $w(a)u \neq 0$ and, by (S2), each value of $a \mapsto w(a)u$ has at most 4 preimages; so

$A_u \le 4\sum_{z\neq 0} 2^{\mathrm{wt}(z)} = 4(3^m-1)$. *Cross.* In case (iv), $r \neq 0$ (otherwise $a' \in \{a, a\oplus u\}$ by APN). For fixed a and $r \neq 0$ there are at most two a'; so $\mathrm{Cross}_u \le 2q\sum_{r\neq 0} 3^{m-\mathrm{wt}(r)} = 2q(4^m-3^m)$. □

Proof of Theorem 3.9. By Lemmas 3.13 and 3.14, each $u \notin \{0,\rho\}$ contributes at most $(4q^3 + 2q\,3^m - 4q)/q^4$, and there are $q-2$ such u, so $\mathcal{E}_{\mathcal{G}} \le (1-2/q)(4+2(3/4)^m - 4/q^2) \le 4 + 2(3/4)^m$. Adding the zero row, Lemma 3.10 and Lemma 3.11 gives $\mathcal{E}_B \le 1 + 2(3/4)^m + 1 + 4 + 2(3/4)^m$. For $m \ge 5$, $6 + 4\cdot 243/1024 = 1779/256$. □

Remark 3.15. Theorem 3.9 gives a constant independent of m, p, d, h; it does not by itself give a mixing statement or any security statement. Its use in the conditional mixing analysis is recorded in Section 6.

4. Local zero sums and their certified correction

Throughout this section $p(x) = x^{q-2}$ (with $0 \mapsto 0$), $m \ge 3$, and $d = \varepsilon \oplus K$ is any $\mathbb{F}_2$-affine map with $\ker K = \{0,\rho\}$, $\rho \neq 0$, $\varepsilon \notin \operatorname{Im} K$ (condition (P2) of Assumption 2.3). Oddness of m is not used in Subsections 4.2–4.4; it enters only when the inverse map is required to be APN, i.e. when the results are combined with Section 3 (Remark 4.10).

4.1. **The local sum.** For an $\mathbb{F}_2$-linear map $L : \mathbb{F}_q \to \mathbb{F}_q$, $\mu \in \mathbb{F}_q$ and $c \in \mathbb{F}_q$ put

$$\Psi_{L,\mu}(c) = \sum_{a\in\mathbb{F}_q} p\big(h(a) \oplus d(a)(c \oplus L(a)) \oplus \mu\big).$$

Proposition 4.1 (Identification with the local coset sum of S_2)**.** *Let $F = F_{p,d,h}$ with (p,d,h) as above, $S_2(x;K_1,K_2) = F(B(B(x)+K_1)+K_2)$, $K_1 = (\kappa,\lambda)$, $K_2 = (\mu,\nu)$, and let $\mathrm{lo}\,s(a)$ be the exclusive or of the low digits of $S_2((a,b);K_1,K_2)$ over $b \in \mathbb{F}_q$ (equivalently the low digit of the ring sum over the coset $(a,*)$). Then, with $L(u) = u \star \kappa$ and $c = p(a) \oplus \lambda \oplus \kappa$, $\mathrm{lo}\,s(a) = \Psi_{L,\mu}(c)$. The key word ν does not occur, λ only translates c, and $a \mapsto c$ is bijective; hence $\#\{a : \mathrm{lo}\,s(a) \neq 0\} = \#\{c : \Psi_{L,\mu}(c) \neq 0\}$.*

Proof. $B(a,b) = (h(a) \oplus d(a)b,\ p(a))$. Adding K_1 (Lemma 2.1) gives low digit $u = h(a) \oplus d(a)b \oplus \kappa$ and high digit $p(a) \oplus \lambda \oplus ((u\oplus\kappa)\star\kappa) = c \oplus L(u)$. Since $d(a) \neq 0$ (Lemma 2.4), $b \mapsto u$ is a bijection of $\mathbb{F}_q$, so the exclusive or over b is an exclusive or over u. The low digit of $B(u, c\oplus L(u)) + K_2$ is $h(u) \oplus d(u)(c \oplus L(u)) \oplus \mu$ (ν enters only the high digit), and the low digit of the final F is p of it. Summing over u gives $\Psi_{L,\mu}(c)$. Finally p is a permutation, so $a \mapsto c$ is bijective. □

Thus $\Psi_{L,\mu} \not\equiv 0$ means that the local zero sum fails on at least one coset, and a lower bound on $\#\{c : \Psi \neq 0\}$ is a lower bound on the number of cosets on which it fails. The converse direction (non-vanishing on every coset) is not claimed.

4.2. **Normalization of d.**

Lemma 4.2. *There is a unique $\eta \in \mathbb{F}_q^*$ with $\operatorname{Im} d = \{z : \operatorname{Tr}(\eta z) = 1\}$. Put $\beta = \eta^{-1}$ and $r(a) = (\eta d(a))^{q/2}$, the square root of $\eta d(a)$. Then r is $\mathbb{F}_2$-affine, $d = \beta r^2$, $r(a \oplus \rho) = r(a)$, and r maps $\mathbb{F}_q$ two-to-one onto $H = \{x : \operatorname{Tr}(x) = 1\}$; the fibre of r over each $x \in H$ is a pair $\{a, a \oplus \rho\}$. Moreover $0 \notin H$.*

Proof. $\operatorname{Im} K$ is an $\mathbb{F}_2$-hyperplane, so $\operatorname{Im} K = \{z : \operatorname{Tr}(\eta z) = 0\}$ for a unique $\eta \neq 0$ (non-degeneracy of the trace form). $\operatorname{Im} d = \varepsilon \oplus \operatorname{Im} K$ is the other coset because $\varepsilon \notin \operatorname{Im} K$, i.e. $\operatorname{Tr}(\eta z) = 1$ on $\operatorname{Im} d$; uniqueness of η follows since a coset of a hyperplane determines the hyperplane and its defining functional. Squaring is $\mathbb{F}_2$-linear and bijective, so r is affine and $r^2 = \eta d$. $\operatorname{Tr}(r(a)) = \operatorname{Tr}(r(a)^2) = \operatorname{Tr}(\eta d(a)) = 1$, so $r(\mathbb{F}_q) \subseteq H$; d is two-to-one with fibres $\{a, a \oplus \rho\}$ (Lemma 2.4), hence so is r, and $|r(\mathbb{F}_q)| = q/2 = |H|$. $\operatorname{Tr}(0) = 0$ gives $0 \notin H$. □

For $d = d_0 = (a^2+a+1)^2$ and odd m one has $\eta = \beta = 1$, $r(a) = a^2+a+1$, $\rho = 1$ (the setting of [2]). Lemma 4.2 uses no change of basis: r is defined from d and the carry $\star$ is untouched.

Lemma 4.3 (Power sums over H; known)**.** *For $m \geq 3$, $p_j := \sum_{x \in H} x^{-j}$ satisfies $(p_1, p_2, p_3, p_4, p_5) = (1, 1, 0, 1, 0)$.*

Proof. $Q(X) = 1 + \sum_{i<m} X^{2^i}$ has $Q' = 1$, $Q(0) = 1$, $\deg Q = q/2$ and root set H, so $Q(X) = \prod_{x \in H}(1 + X/x)$ and the elementary symmetric functions of $\{1/x\}$ are $e_1 = e_2 = e_4 = 1$, $e_3 = e_5 = 0$ ($e_5 = 0$ also for $m = 3$, where $\deg Q = 4$). Newton's identities in characteristic two give $p_1 = 1$, $p_2 = p_1^2 = 1$, $p_3 = e_1p_2 + e_2p_1 + e_3 = 0$, $p_4 = p_2^2 = 1$, $p_5 = e_1p_4 + e_2p_3 + e_3p_2 + e_4p_1 + e_5 = 0$. □

4.3. **The linear certificate.** For $x = r(a) \in H$ put $\Delta_h(x) = h(a) \oplus h(a \oplus \rho)$ (independent of the choice in the fibre) and

$$A_d(h) = \sum_{x \in H} x^{-3} \Delta_h(x), \qquad B_d(h) = \sum_{x \in H} x^{-5} \Delta_h(x)^2,$$

$$\ell_d(h) = A_d(h) + B_d(h)^{q/2}, \qquad \tau = L(\rho).$$

Let $s = q/2-1$, $z(a) = h(a) \oplus d(a)L(a) \oplus \mu$ and $\Phi(C) = \sum_a (z(a)+d(a)C)^s \in \mathbb{F}_q[C]$.

Theorem 4.4 (Top coefficients)**.** *For every h, L, μ: $\Psi_{L,\mu}(c) = \Phi(c)^2$ for all $c \in \mathbb{F}_q$, and*

$$[C^s]\Phi = 0, \qquad [C^{s-1}]\Phi = \beta^{s-1}\big(A_d(h) + \beta\tau\big),$$
$$[C^{s-2}]\Phi = \beta^{s-2}\big(B_d(h) + \beta^2\tau^2\big).$$

Proof. $q - 2 = 2s$ and Frobenius give $\Psi = \Phi^2$ pointwise. The binary expansion of s consists of $m-1$ ones, so $\binom{s}{k} \equiv 1 \pmod 2$ for $0 \leq k \leq s$ (Lucas) and $[C^k]\Phi = \sum_a z(a)^{s-k} d(a)^k$. Group a in fibres $\{a, a \oplus \rho\}$ of d: d is constant

on a fibre, so $[C^s]\Phi = \sum_a d(a)^s = 0$. On a fibre, $L(a \oplus \rho) = L(a) \oplus \tau$, hence $z(a) \oplus z(a \oplus \rho) = \Delta_h(x) \oplus \beta x^2 \tau$ with $x = r(a)$ (μ cancels). With $d = \beta x^2$ and $x^{q-1} = 1$: $d^{s-1} = \beta^{s-1} x^{-3}$, $d^{s-2} = \beta^{s-2} x^{-5}$ (here $s \geq 3$ for $m \geq 3$; for $m = 3$, $s - 2 = 1$). Therefore $[C^{s-1}]\Phi = \beta^{s-1} \sum_H x^{-3}(\Delta_h + \beta x^2 \tau) = \beta^{s-1}(A_d + \beta \tau p_1)$ and $[C^{s-2}]\Phi = \beta^{s-2} \sum_H x^{-5}(z(a)^2 \oplus z(a \oplus \rho)^2) = \beta^{s-2} \sum_H x^{-5}(\Delta_h + \beta x^2 \tau)^2 = \beta^{s-2}(B_d + \beta^2 \tau^2 p_1)$, and $p_1 = 1$ (Lemma 4.3). □

Theorem 4.5 (Certificate)**.** *If $\ell_d(h) \neq 0$ then for every $\mathbb{F}_2$-linear L and every μ, $\Phi \neq 0$, $\deg \Phi \leq q/2 - 2$ and*

$$\#\{c \in \mathbb{F}_q : \Psi_{L,\mu}(c) \neq 0\} \;\geq\; q/2 + 2.$$

More precisely $\deg \Phi = q/2 - 2$ if $\tau \neq \eta A_d(h)$, and $\deg \Phi = q/2 - 3$ (so at least $q/2 + 3$ non-zero values) if $\tau = \eta A_d(h)$. By Proposition 4.1, the local zero sum of S_2 then fails on at least $q/2 + 2$ cosets for every local key.

Proof. $\ell_d(h)^2 = A_d^2 + B_d$. If both coefficients of Theorem 4.4 vanished, then $A_d = \beta\tau$ and $B_d = \beta^2\tau^2 = A_d^2$, i.e. $\ell_d(h) = 0$. So $\Phi \neq 0$ and, since $[C^s]\Phi = 0$, $\deg \Phi \leq s - 1$; a non-zero polynomial of degree $\leq s - 1$ has at most $s - 1$ roots, and $\Psi(c) = 0 \iff \Phi(c) = 0$. If $\tau = \eta A_d$ then the first coefficient vanishes and the second is $\beta^{s-2}(B_d + A_d^2) = \beta^{s-2}\ell_d^2 \neq 0$. □

The condition $\ell_d(h) \neq 0$ is sufficient; its necessity is not asserted. Theorem 4.5 quantifies over *all* $\mathbb{F}_2$-linear L, whereas the local keys of Proposition 4.1 only produce the carry-mask maps $L(a) = a \star \kappa$; the two quantifiers are kept apart in the examples below (Examples 4.12 and 4.13).

Theorem 4.6 (Linearity, explicit correction, exact pass fraction)**.** *Let $\chi : \mathbb{F}_q \to \mathbb{F}_2$ be $\mathbb{F}_2$-linear with $\chi(\rho) = 1$ and $g_d(a) = \chi(a)\, r(a)$. Then*

1. *$\ell_d : \mathbb{F}_q^{\mathbb{F}_q} \to \mathbb{F}_q$ is $\mathbb{F}_q$-linear and $\ell_d(g_d) = 1$; in particular ℓ_d is surjective;*
2. *$\ell_d(h + \lambda g_d) = \ell_d(h) + \lambda$ for all h and $\lambda \in \mathbb{F}_q$; exactly one of the q functions $h + \lambda g_d$ fails the certificate, and $h^\sharp = h$ ($\ell_d(h) \neq 0$), $h^\sharp = h + g_d$ ($\ell_d(h) = 0$) always passes;*
3. *for every $\mathbb{F}_q$-linear subspace $V \subseteq \mathbb{F}_q^{\mathbb{F}_q}$ with $g_d \in V$, exactly a fraction $1 - 1/q$ of V passes the certificate; this applies to $V = \mathbb{F}_q^{\mathbb{F}_q}$ and to the space $\mathcal{Q}_2$ of functions of Boolean degree ≤ 2 (in a fixed basis), $\dim_{\mathbb{F}_q} \mathcal{Q}_2 = 1 + m + \binom{m}{2}$.*

Proof. (1) Δ_h is $\mathbb{F}_q$-linear in h; $x \mapsto x^{q/2}$ is additive with $(xy^2)^{q/2} = x^{q/2}y$, so $B_d(h)^{q/2} = \sum_H (x^{-5})^{q/2} \Delta_h(x)$ is $\mathbb{F}_q$-linear in h. Since $r(a \oplus \rho) = r(a)$ and $\chi(a \oplus \rho) = \chi(a) \oplus 1$, $\Delta_{g_d}(x) = x$; hence $A_d(g_d) = p_2 = 1$, $B_d(g_d) = p_3 = 0$, $\ell_d(g_d) = 1$. (2) is linearity. (3) $\ell_d|_V$ is a surjective linear functional, so its kernel has index q in V. Each coordinate of g_d is $\chi \cdot r_i$, a product of two affine Boolean functions, so $g_d \in \mathcal{Q}_2$; a basis of $\mathcal{Q}_2$ over $\mathbb{F}_q$ is given by the square-free monomials of degree ≤ 2 in the coordinates of a. □

The fraction in (3) is the fraction passing a sufficient certificate, not the fraction of h without local zero sums, and not a fraction of ciphers that are secure.

4.4. **Boolean degree two is minimal for this local problem.**

Proposition 4.7. *If h is $\mathbb{F}_2$-affine then $\Psi_{0,\mu}(c) = 0$ for all c, μ. Consequently every h with $\Psi_{L,\mu} \not\equiv 0$ for all (L, μ) (in particular g_d and every h passing the certificate) has Boolean degree ≥ 2, and g_d has Boolean degree exactly 2.*

Proof. For $L = 0$ and fixed c, μ, $a \mapsto y(a) = h(a) \oplus d(a)c \oplus \mu$ is $\mathbb{F}_2$-affine. If it is not injective, every fibre is a coset of its kernel and has even size, so $\sum_a p(y(a)) = 0$. If it is bijective, $\sum_a p(y(a)) = \sum_{x \in \mathbb{F}_q} x^{q-2} = \sum_x x = 0$ ($m \geq 2$). By Theorem 4.5, $\ell_d(g_d) = 1$ forces $\Psi_{0,\mu} \not\equiv 0$, so g_d is not affine. □

This is a statement about the Boolean degree of h for this local problem only; it says nothing about AND counts, about the degree of F or of the whole arrangement, or about optimality among known constructions.

4.5. **Evaluation and the unchanged differential guarantees.**

Remark 4.8 (Cost of the correction)**.** The coordinates of r are public affine forms in the bits of a and $\chi(a)$ is one public linear form, so g_d is evaluated with at most m AND gates plus linear gates (in fact $m-1$, Proposition 4.9); λg_d for a fixed λ adds only an $\mathbb{F}_2$-linear map (no AND). This bounds the cost of the correction, not of h itself. The quantities η, ρ, χ are fixed off line; computing $\ell_d(h)$ for an arbitrary table h may require q evaluations and is not claimed to be cheap for large m. Choosing $h = g_d$ from the start needs no scan of h.

Proposition 4.9 (Multiplicative complexity of the correction alone)**.** *In the model of XOR/NOT/AND straight-line circuits with two-input AND gates counted and linear gates free, the multiplicative complexity of g_d (as a map $\mathbb{F}_2^m \to \mathbb{F}_2^m$) is exactly $m - 1$.*

Proof. Choose an $\mathbb{F}_2$-linear isomorphism O of $\mathbb{F}_q$ whose first coordinate is Tr. Since $r(\mathbb{F}_q) = H$, $Or(a) = (1, v_1(a), \ldots, v_{m-1}(a))$ with v_i affine. The map $a \mapsto (t, v_1, \ldots, v_{m-1})$, $t = \chi(a)$, is an affine bijection: if two points have the same v, their difference lies in $\{0, \rho\}$ (the fibres of r), and $\chi(\rho) = 1$ separates them. In these coordinates $Og_d = (t, tv_1, \ldots, tv_{m-1})$, which uses $m - 1$ AND gates. Conversely, in a circuit with k AND gates every output is an affine combination of the inputs and of the k AND outputs, so modulo affine functions the outputs span a space of dimension at most k. The functions $tv_1, \ldots, tv_{m-1}$ are linearly independent modulo affine functions (distinct square-free quadratic monomials in the coordinates (t, v)), and an affine change of the input and a linear change of the output do not change this dimension; hence $k \geq m - 1$. □

This concerns g_d computed on its own. It is not a bound for F, nor for $h + g_d$, nor for the additional cost when the correction shares intermediate values with the inverse map (for $h = a^3$ no additional AND gate is needed). The upper bound and the quotient-dimension lower bound are standard principles of multiplicative complexity; only the identification of the normal form is specific to g_d.

Remark 4.10 (Same differential guarantees)**.** The correction changes h only. For odd $m \geq 5$ the inverse map is an APN permutation, so $(p, d, h^\sharp)$ satisfies Assumption 2.3 and Theorems 3.4, 3.7 (constant $4/N$ when $\mathrm{wt}(\rho) = 1$, $(6 + 6/q)/N$ in general), 3.9 and 5.3 apply to it verbatim, with the same independence and reset assumptions. For even m only the algebra of Subsections 4.2–4.4 holds.

Corollary 4.11 (The case $d = d_0$)**.** *For odd $m \geq 3$ and $d = d_0$: $\eta = \beta = 1$, $\rho = 1$, and Theorems 4.4–4.6 reduce to $[C^{s-1}]\Phi = A + \tau$, $[C^{s-2}]\Phi = B + \tau^2$, $\ell = A + \sqrt{B}$. For $h(a) = a^3$: $\Delta_h = r$, $A = 1$, $B = 0$, $\ell = 1$, so $\Psi_{L,\mu} \not\equiv 0$ for every local key, with at least $q/2 + 2$ non-zero values if $\tau \neq 1$ and at least $q/2 + 3$ if $\tau = 1$. The correction $h + \lambda a^3$ and the correction $h + \lambda g_{d_0}$ have the same effect on ℓ ($\ell(a^3) = \ell(g_{d_0}) = 1$).*

The function $h = a^3$ remains the main comparison point: it is shared with the inverse chain and its certificate value is 1; the general construction does not replace it.

Example 4.12 (Vanishing; non-necessity)**.** For $m = 5$, $d = d_0$, $h(a) = a^{21}$ and $L = 0$, $\mu = 0$: $\Psi_{0,0} \equiv 0$ on $\mathbb{F}_{32}$ (substitute $a = b^3$; the expansion of $(X + C(X^{12} + X^6 + 1))^{30}$ has 128 monomials and no positive X-degree divisible by 31). Here $\ell(h) = 0$. Conversely, for random h with $\ell_d(h) = 0$ we observed no *carry-mask* key $(L, \mu) = (a \mapsto a \star \kappa, \mu)$ with $\Psi \equiv 0$ (Section 7); so, for the carry-mask key family of this fixed model, $\ell_d = 0$ is not a vanishing criterion. Neither observation is a classification.

Example 4.13 (The two quantifiers differ)**.** Let $m = 3$, $\mathbb{F}_8 = \mathbb{F}_2[t]/(t^3 + t + 1)$ with the polynomial basis, $d = d_0$, and let h have the value table $(h(0), \ldots, h(7)) = (7, 5, 2, 2, 2, 0, 7, 2)$ (index $a = a_0 + 2a_1 + 4a_2$). Then $\ell(h) = 0$. For each of the 64 carry-mask keys (κ, μ) the function $\Psi_{L,\mu}$ with $L(a) = a \star \kappa$ is not identically zero; but for the $\mathbb{F}_2$-linear map L with $(L(1), L(t), L(t^2)) = (3, 0, 0)$ and $\mu = 0$ one has $\Psi_{L,0} \equiv 0$ (exact computation; among all $512 \cdot 8$ pairs (L, μ) with L linear, 64 give $\Psi \equiv 0$). So a finite check over carry-mask keys says nothing about non-necessity for general L, and the statement after Theorem 4.5 is restricted accordingly. This example is not a counterexample to Theorem 4.5, which is a sufficient condition.

5. Composition: an averaged bound and what it does not prevent

This section concerns the arrangement of Section 2.6, with every local layer built from one table $F = F_{p,d,h}$ satisfying Assumption 2.3, $m \geq 5$ odd.

Everything here is an application of known principles; we state the results in the form needed and give the short proofs to keep the paper self-contained, without claiming novelty for the principles.

5.1. **Two standard lemmas.**

Lemma 5.1 (Post-composition with a reset key; standard)**.** *Let G be a finite abelian group, $E_\kappa : G \to G$ a family of maps, M an automorphism of G, and $C_k : G \to G$ a family of permutations. Let κ, L, k be independent with L uniform on G. Then*

$$\mathbb{E}_{\kappa,L,k}\mathrm{DP}_{C_k\circ\tau_L\circ M\circ E_\kappa}(\alpha,\gamma) = \sum_{\delta} \mathbb{E}_\kappa \mathrm{DP}_{E_\kappa}(\alpha,\delta)\, \bar{T}_C(M\delta,\gamma),$$

with $\bar{T}_C = \mathbb{E}_k T_{C_k}$, where $\tau_L(y) = y + L$. In particular, if $\mathbb{E}_\kappa \mathrm{DP}_{E_\kappa}(\alpha,\delta) \le b_\star$ for all δ, then the left side is at most $b_\star$.

Proof. For a fixed pair (y, y') with $y' - y = M\delta$, the number of L with $C(y' + L) - C(y + L) = \gamma$ is $\delta_C(M\delta,\gamma)$ (translation). Averaging over the inputs and κ gives the identity. Each C_k is a permutation, so $\bar{T}_C$ has column sums 1, and since M is bijective, $\sum_\delta \bar{T}_C(M\delta,\gamma) = 1$; the bound follows. (Without the reset L the product formula fails in general.) □

Lemma 5.2 (Two-layer MDS count for averaged kernels; standard)**.** *Let R be a finite commutative ring, $M \in \mathrm{Mat}_w(R)$ with all minors units, and let T_i, U_j ($1 \le i, j \le w$) be non-negative $R \times R$ matrices with row sums 1, $T(0,\cdot) = e_0$, $T(\alpha, 0) = 0$ for $\alpha \ne 0$, and all entries of non-zero rows at most $b_\star$. Put $\mathsf{K}(\alpha,\gamma) = \sum_{\delta\in R^w} \prod_i T_i(\alpha_i,\delta_i) \prod_j U_j((M\delta)_j,\gamma_j)$. Then $\mathsf{K}(\alpha,\gamma) \le b_\star^w$ for every $\alpha \ne 0$ and every γ.*

Proof. Let $I = \operatorname{supp}\alpha$, $J = \operatorname{supp}\gamma$, $a = |I| \ge 1$, $b = |J|$. Non-zero terms have $\operatorname{supp}\delta = I$ and $\operatorname{supp} M\delta = J$. If $a + b \le w$ there is no such $\delta \ne 0$ (branch number $w + 1$), so $\mathsf{K} = 0$. If $a + b > w$, choose $I' \subseteq I$ with $|I'| = a + b - w$; the condition $(M\delta)_{J^c} = 0$ determines $\delta_{I\setminus I'}$ from $\delta_{I'}$ because the $(w-b)\times(w-b)$ minor $M_{J^c, I\setminus I'}$ is invertible. Bound the $w - b$ factors T_i, $i \in I \setminus I'$, and the b factors U_j, $j \in J$, by $b_\star$, and sum the remaining $\prod_{i\in I'} T_i(\alpha_i,\delta_i)$ over $\delta_{I'}$ to at most 1. □

5.2. **The averaged multi-word bound.**

Theorem 5.3. *Let $q \ge 32$, $w \ge 2$, M_1 with all minors units and M_2, M_3 invertible, all key words independent and uniform. For every h and every $\alpha \ne 0$, γ:*

1. *if $\mathrm{wt}(\rho) = 1$, then $\mathbb{E}_\mathcal{K}\mathrm{DP}_E(\alpha,\gamma) \le (4/N)^w$;*
2. *for general ρ, $\mathbb{E}_\mathcal{K}\mathrm{DP}_E(\alpha,\gamma) \le ((6 + 6/q)/N)^w \le (99/(16N))^w$.*

Proof. Let $b_\star$ be the bound of Theorem 3.7 in the relevant case. By Proposition 2.6 (with the whitening k_0, resp. the reset L_1, providing the uniform input), the averaged kernel of each word of $S_2^{(1)}$ and of $S_2^{(2)}$ is $T_B^2 T_F$, which satisfies the hypotheses of Lemma 5.2 with $b_\star$ (it is the averaged kernel of

permutations). With L_1 uniform and keys independent across words, the averaged kernel of $x \mapsto X_2$ is the kernel K of Lemma 5.2 for $M = M_1$ ([5, 6]), hence at most $b_\star^w$. Apply Lemma 5.1 with $(M_2, L_2, S_1^{(3)})$ and then with $(M_3, L_3, S_1^{(4)})$; each $S_1(\cdot; k)$ is a permutation. □

Remark 5.4. The bound is an average over the $10w$ independent key words. It is not a fixed-key bound, it says nothing about the key schedule, and the reset keys L_2, L_3 are essential to the proof. M_1 uses the MDS property; M_2, M_3 only invertibility.

5.3. **A candidate-existence distinguisher.**

Lemma 5.5 (Zero sum of all ring elements; known)**.** $\sum_{r\in R} r = 0$ *for* $R = \mathrm{GR}(4, m)$*,* $m \geq 2$*, and* $N \cdot y = 0$ *for every* $y \in R$*.*

Proof. $(R, +) \cong (\mathbb{Z}/4)^m$; in each coordinate each residue occurs 4^{m-1} times and $4^{m-1}(0 + 1 + 2 + 3) \equiv 0 \pmod 4$ for $m \geq 2$. Also $4 \mid N$. □

Lemma 5.6 (Zero sum at V_2)**.** *Fix all keys and let* x *range over a one-word slice* $\{x : x_a \in R,\ x_j = c_j\ (j \neq a)\}$*. Then* $\sum_x V_2(x) = 0$ *in* R^w*.*

Proof. $X_{1,a}$ runs over R bijectively and the other words of X_1 are constant. Since $M_{1,ia}$ is a unit, $V_{1,i}$ is a bijective function of x_a for every i, hence so is $X_{2,i}$. By Lemma 5.5, $\sum_x X_{2,i} = 0$, and $\sum_x V_{2,j} = \sum_i M_{2,ji} \cdot 0 + NL_{2,j} = 0$. □

For a candidate $\theta = (K_4, c, K_3) \in R^w \times R^w \times R^w$ define the word-wise map

$$G_\theta(y) = S_1^{-1}\big(M_3^{-1} S_1^{-1}(y; K_4) - c;\ K_3\big).$$

G_θ is a bijection of R^w, and for $\theta^* = (K_4, M_3^{-1}L_3, K_3)$ one has $G_{\theta^*}(E(x)) = V_2(x)$. The distinguisher D_s queries s pairwise disjoint one-word slices $X_1, \ldots, X_s$ (fixed in advance, non-adaptively; $Q = sN$ distinct inputs) and accepts iff

$$A: \quad \exists\theta\ \forall t \leq s\ \forall j:\ \textstyle\sum_{x\in X_t} G_\theta(y(x))_j = 0.$$

Theorem 5.7 (Candidate-existence distinguisher)**.** *Let* $M = N^w$ *and* $Q = sN \leq M$*.*

1. *On the arrangement* E*,* D_s *accepts with probability* 1 *for every key.*
2. *For a uniformly random function* RF*,* $\Pr[A] \leq N^{(3-s)w}$*.*
3. *For a uniformly random permutation* RP*,* $\Pr[A] \leq \min\{1,\ N^{(3-s)w} + \binom{Q}{2}/M\}$ *and*

$$\Pr_{\mathrm{RP}}[A] \leq \min\Big\{1,\ N^{(3-s)w}\,\frac{M^Q}{(M)_Q}\Big\}, \qquad (M)_Q = M(M-1)\cdots(M-Q+1).$$

No independence between different candidates, or between output words, is used.

Proof. (1) Lemma 5.6 and θ^*. (2) Fix θ. The answers of RF on the Q distinct inputs are independent and uniform in R^w, and G_θ is a bijection, so the values $G_\theta(y(x))$ are independent and uniform. The sum over one slice of N independent uniform elements is uniform, so it vanishes with probability N^{-w}; disjoint slices give independent sums, hence probability N^{-sw}. A union bound over the N^{3w} candidates gives the claim. (3) On fixed distinct inputs, the output distribution of RP equals that of RF conditioned on the event D that all Q outputs are distinct, and $\Pr_{\mathrm{RF}}[\mathsf{D}] = (M)_Q/M^Q$. Hence $\Pr_{\mathrm{RP}}[A] = \Pr_{\mathrm{RF}}[A \cap \mathsf{D}]/\Pr_{\mathrm{RF}}[\mathsf{D}] \le \Pr_{\mathrm{RF}}[A]\, M^Q/(M)_Q$; no independence of A and D is used. The additive form follows from $|\Pr_{\mathrm{RP}}[A] - \Pr_{\mathrm{RF}}[A]| \le \Pr_{\mathrm{RF}}[\overline{\mathsf{D}}] \le \binom{Q}{2}/M$. □

Proposition 5.8 (Decomposition of the acceptance condition)**.** *$G_\theta(y)_j$ depends only on K_4, c_j and $K_{3,j}$. Hence A holds iff $\exists K_4 \bigwedge_j \exists (c_j, K_{3,j}) : P_j(K_4, c_j, K_{3,j})$, where P_j is the slice condition for word j. An implementation that, for each K_4, searches each target word separately computes D_s exactly; it examines $N^w \cdot N^2$ cells per target word.*

Proof. For fixed K_4, the conditions for different j involve disjoint variables $(c_j, K_{3,j})$. □

5.4. **Numbers, resources and recovered objects.** Table 2 lists the false-acceptance bounds for RP (exact rational arithmetic). For two words the additive bound is void, while the conditioned bound is not.

TABLE 2. Bounds on $\Pr_{\mathrm{RP}}[A]$ from Theorem 5.7(3).

case	Q	$\log_2$ cond.	cond. (outward)	additive
$m=3, w=2, s=4$	256	-0.2567	$< 9/10$	≈ 7.97 (void)
$m=3, w=2, s=5$	320	-5.536	$< 1/40$	≈ 12.46 (void)
$m=5, w=2, s=4$	4096	-8.446	$< 1/300$	≈ 7.998 (void)
$m=5, w=2, s=5$	5120	-21.94	$< 1/4\,000\,000$	≈ 12.50 (void)
$m=5, w=4, s=4$	4096	-39.99999	$< 2/N^4$	$8386561/2^{40}$

Table 3 fixes, for each experiment, the game, what is unknown, what is recovered, and the resources; each entry is marked as a theorem, a finite measurement, or an extrapolation. Data is $D = sN$ chosen plaintexts; T is counted in local multiply-accumulate operations of the implemented enumeration; this is the cost of one implementation, not a lower bound.

5.5. **A small averaged differential bound is not PRP security.** The same arrangement satisfies Theorem 5.3 and Theorem 5.7. For $m = 5$, $w = 4$ the averaged differential probability is at most $(4/N)^4 = 2^{-32}$ for every non-zero input difference, while D_4 uses $4N = 2^{12}$ chosen plaintexts and distinguishes E from a random permutation with advantage at least $1 - 2^{-39.99}$. There is no contradiction: Theorem 5.3 is an average over independent keys of a differential quantity, whereas D_s exploits an integral

TABLE 3. Resources and recovered objects (the recovered object is at most θ, i.e. $3w$ of the $10w$ key words).

experiment	unknown	recovered	$T/D/M$	outcome / status
full-blind, $m = 3, w = 2$	all of θ	θ	0.77 s, $1.54 \cdot 10^{10}$ ops; $D = 4N = 256$; 21 MB	$s \geq 4$: unique candidate $=$ true θ in 144/144 runs; $s = 3$ ambiguous in 33/48 (verified, finite)
full-blind, $m = 5, w = 2$	all of θ	—	extrapolated $2.7 \cdot 10^{16}$ ops (72–87 h single process); partial-sum variant $5.6 \cdot 10^{15}$ ops, $\approx$ 30 h (Poisson-type extrapolation from $m = 5$ timings: neither a measurement nor a bound)	not run
candidate existence, $m = 5, w = 4, s = 4$	existence of θ	none (distinguishing only)	$D = 4N$; implemented $T \approx 5.1 \cdot 10^{22}$ ops	advantage $\geq 1 - 2^{-39.99}$ (Theorem 5.7)

property (Lemma 5.6) and a candidate set of size N^{3w}, far smaller than the key space N^{10w}. The implemented enumeration for $m = 5$, $w = 4$ costs more than the codebook N^w; we prove no lower bound on the time of any attack, and we do not claim that no attack below the codebook exists.

The local correction of Section 4 acts inside one S_2 block. It removes the local zero sum of Proposition 4.1 for every local key, but it does not remove the word-level zero sum of Lemma 5.6, which holds for every bijective local layer. Consistently, in the finite experiments the tables compared (natural, $h = 0$, one further fixed table, $h = a^3$) behaved alike: in the two-layer oracle test all four kept only the true local candidates (16/16 targets), the $m = 3$, $w = 2$ full-blind runs with $h \in \{0, a^3\}$ were unique for $s \geq 4$, and the SAT experiments on the same arrangement did not separate the tables. With the correction itself ($m = 3$, $w = 2$, $s = 4$, the same 16 keys and slices for eight tables including $h = a^3$, h_0, g_{d_0} and $h_0 + g_{d_0}$, the latter changing ℓ from 0 to 1 and the number of local keys with $\Psi \equiv 0$ from 32 to 0), the real construction was accepted in 16/16 runs for every table, θ was recovered

uniquely in 127/128 runs, and RF, RP were accepted in 0/16 runs each (finite measurement; the case $m = 5$ was not run). A local certificate and a word-level candidate-existence distinguisher are different statements; the success of the correction does not neutralize the distinguisher.

6. Mixing of the alternating arrangement: a uniform one-word contraction

This section concerns the averaged (Markov) transition of the alternating two-word arrangement of [2]-type blocks with an MDS mixing M whose entries and determinant are units. Its statements are about key-averaged transitions only. Theorem 6.2 gives a contraction exponent that is positive and uniform over all PD data and all h; with it, Theorem 6.9 reaches the scale $1/N$ in a fixed number of steps without further hypotheses. For the inverse map $p(x) = x^{q-2}$ with m odd and $d = 1 \oplus K_0(x/\rho)$, $K_0(X) = X^4 + X^2$, $\rho \neq 0$, the exponent is $1/2$ (Corollary 6.6), the largest possible for any PD data (Proposition 6.3); for the inverse map with a general admissible d the constant obtained grows with the linearized degree of d, and the exponent $1/2$ uniformly over all PD data is open (Remark 6.7). The constants are explicit but weak for small m.

6.1. **One word: the partial Fourier picture.** Let $T = T_B$ be the averaged transition matrix of B on ring differences ($N \times N$, doubly stochastic, $T(\alpha, 0) = [\alpha = 0]$). For a low digit u of the input difference and s of the output difference, use on each fibre the orthonormal Walsh basis in the high digit:

$$\widehat{T}((u,\eta),(s,\xi)) = q^{-1} \sum_{v,t} (-1)^{\langle \eta, v\rangle + \langle \xi, t\rangle}\, T((u,v),(s,t)),$$

where $\langle \cdot, \cdot \rangle$ is the coordinate inner product of the fixed basis. For $u \notin \{0, \rho\}$ put $k = K(u)$, $A_a = d(a)d(a \oplus u)/k$, $D_u p(a) = p(a \oplus u) \oplus p(a)$ and $C_a(s) = h(a) \oplus \frac{d(a)}{k}\big[s \oplus h(a \oplus u) \oplus h(a) \oplus d(a \oplus u)(a \star u)\big]$.

Proposition 6.1 (Partial Fourier formulas)**.** *For every (p, d, h) under Assumption 2.3:*

1. *($\mathcal{G}$)* $\widehat{T}((u,\eta),(s,\xi)) = q^{-2} \sum_{a:\, M^*_{A_a}(s\star\xi)=\eta} (-1)^{\langle \xi, D_u p(a)\rangle + \langle s\star\xi,\, C_a(s)\rangle}$, *where $M_A : x \mapsto Ax$ and M^*_A is its adjoint for $\langle \cdot, \cdot \rangle$. For $\eta \neq 0$ the sum is either empty or runs over one coset $a \oplus \langle \rho, u \rangle$ (four points), and the four signs multiply to $+1$; hence every such entry lies in $\{0, \pm 4/q^2\}$.*
2. *($\mathcal{E}$, $u = \rho$)* $\widehat{T} = q^{-2}[s \star \xi = 0] \sum_a (-1)^{\langle \eta, v_a(s)\rangle + \langle \xi, D_\rho p(a)\rangle}$ *with* $v_a(s) = (s \oplus h(a) \oplus h(a \oplus \rho))/d(a) \oplus (a \star \rho)$.
3. *($\mathcal{Z}$, $u = 0$)* $\widehat{T} = q^{-2}[s \star \xi = 0] \sum_a (-1)^{\langle \eta, s/d(a)\rangle}$.

Proof. By Lemma 3.2, for each a the high input digit b is determined in the $\mathcal{G}$ case by the low equation, and the high output difference is $t = D_u p(a) \oplus s \star (A_a v \oplus C_a(s))$; summing the characters over t and then over v leaves the

indicator of $\eta = M_{A_a}^*(s \star \xi)$. For fixed $z = s \star \xi \neq 0$ the map $A \mapsto M_A^* z$ is injective ($\langle z, Ax\rangle = \mathrm{Tr}(\omega Ax)$ for some $\omega \neq 0$), so A_a is determined; A_a is constant on $a \oplus \langle \rho, u\rangle$ and the two roots $\{y, y \oplus k\}$ of $y^2 + ky + kA = 0$ have d-preimages forming one such coset. On that coset $D_u p$ takes one value twice in each of two pairs, and the four values of $C_a(s)$ sum to 0 (the h-terms cancel because $d(a)/k + d(a \oplus u)/k = 1$, the s-terms and the $\star$-terms appear with equal coefficients an even number of times). In the $\mathcal{E}$ and $\mathcal{Z}$ cases the low equation does not involve b; the sum over b of $(-1)^{\langle s\star\xi,\, h(a)\oplus d(a)b\rangle}$ equals $q[s \star \xi = 0]$, and the low equation determines v from (a, s). □

The entrywise bound $4/q^2$ is not an operator-norm bound; the next theorem bounds the operator norm.

6.2. **The uniform contraction R1.** Let $W = R \setminus \{0\}$, $T' = T|_{W\times W}$, and let $\perp \subset \ell^2(W)$ be the functions whose sum over every low-digit fibre $f_s \cap W$ vanishes; $\tau := \|\Pi_\perp T'^{\top} \Pi_\perp\|$ (the one-word fluctuation contraction used in the two-word reduction of Appendix A).

Theorem 6.2 (R1)**.** *For every $m \geq 3$, every (p, d, h) under Assumption 2.3:*

$$\tau^2 \;\leq\; \frac{4(1.5^m - 1)}{q} + \frac{3}{q} + \frac{3^m - 2q + 1}{q^2} \;=\; 5\Big(\frac{3}{4}\Big)^m - \frac{3}{q} + \frac{1}{q^2},$$

so that $\tau \leq \sqrt{5}\, q^{-\gamma}$ with $\gamma = 1 - \frac{1}{2}\log_2 3 \approx 0.2075$.

Proof. Blocks. The input space $\perp$ is the orthogonal sum of the fibres $u \neq 0$ (Walsh directions $\eta \neq 0$) and the fibre $u = 0$ (zero-sum functions on $q - 1$ points, extended by 0 at the origin: an isometry into the directions $\eta \neq 0$). Hence $\Pi_\perp T'^{\top} \Pi_\perp = [M_{\mathcal{G}} \,|\, M_{\mathcal{E}} \,|\, M_{\mathcal{Z}}]$ and $\tau^2 \leq \|M_{\mathcal{G}}\|^2 + \|M_{\mathcal{E}}\|^2 + \|M_{\mathcal{Z}}\|^2$. On an output fibre $s \neq 0$, $\perp$ is spanned by $\xi \neq 0$; on the output fibre $s = 0$ ($q - 1$ points), the projection of $f|_W$ onto zero-sum functions has norm at most $\|f - \mathrm{mean}(f)\|$, the norm of the $\xi \neq 0$ part of f. So each $\|M_\bullet\|$ is at most the norm of the corresponding rows $\eta \neq 0$ and columns $\xi \neq 0$ of $\widehat{T}$.

$\mathcal{G}$. Let $g = q^2 \widehat{T}$ on these rows and columns and $Q = gg^{\top}$. For rows $r = (u, \eta)$, $r' = (u', \eta')$, atoms $a \in$ fibre of r and $a' \in$ fibre of r' share the column (s, ξ) iff $s \star \xi = z$, where $z = z_a(\eta)$ is the unique solution of $M_{A_a}^* z = \eta$, and $z_{a'}(\eta') = z$; then $z \neq 0$. Summing over the $2^{m-\mathrm{wt}\, s}$ vectors ξ with $s \star \xi = z$ gives $2^{m-\mathrm{wt}\, s}(-1)^{\langle z,\delta\rangle}[\delta \subseteq s]$ with $\delta = D_u p(a) \oplus D_{u'} p(a')$. The function h enters only through the phase $\langle z, C_a(s) \oplus C'_{a'}(s)\rangle$; taking absolute values and summing over $s \supseteq z \vee \delta$, $\sum_{s \supseteq w} 2^{m-\mathrm{wt}\, s} = 3^{m-\mathrm{wt}\, w}$, gives $|Q[r, r']| \leq \sum_{(a,a')} 3^{m-\mathrm{wt}(z\vee\delta)}$. For fixed r and a, every pair (u', a') determines η', so the absolute row sum is at most $\sum_a \sum_{u',a'} 3^{m-\mathrm{wt}(z_a \vee \delta)}$. As p is a permutation, each value $y = D_{u'} p(a')$ is taken at most q times, so the inner sum is at most $q \sum_{\delta \in \mathbb{F}_q} 3^{m-\mathrm{wt}(z\vee\delta)} = q\, 4^m 2^{-\mathrm{wt}\, z}$. Each $z \neq 0$ arises from at most four a (one coset), so $\sum_a 2^{-\mathrm{wt} z_a} \leq 4((3/2)^m - 1)$. By Schur's test $\|M_{\mathcal{G}}\|^2 \leq \|Q\|/q^4 \leq 4(1.5^m - 1)/q$. ($\mathcal{G}$ rows do not reach the output fibre $s = 0$: there $s \star \xi = 0$ and $\eta \neq 0$ leaves an empty sum.)

$\mathcal{E}$. Write the block as $q^{-2}\Phi N$ with $\Phi[\eta, v] = (-1)^{\langle \eta, v\rangle}$ ($\|\Phi\| \le \sqrt{q}$) and $N[v,(s,\xi)] = [s\star\xi = 0]\sum_{a:v_a(s)=v}(-1)^{\langle \xi, D_\rho p(a)\rangle}$. In $NN^\top[v,v']$ the sum over $\xi \neq 0$ with $s\star\xi = 0$ is at most $2^{m-\mathrm{wt}s}[\delta \subseteq s] + 1$ in absolute value. Since $D_\rho p$ is two-to-one (p APN), for fixed a at most $2^{\mathrm{wt}s+1}$ values a' have $\delta \subseteq s$; so the row sum is at most $\sum_s \#\{a : v_a(s) = v\}\cdot 3q = 3q^2$, because $s \mapsto v_a(s)$ is bijective for each a. Hence $\|M_{\mathcal{E}}\|^2 \le q^{-4}\cdot q\cdot 3q^2 = 3/q$.

$\mathcal{Z}$. Here $s = d(a)v \neq 0$, and the entry does not depend on ξ: for fixed s the block has $n_s = 2^{m-\mathrm{wt}s} - 1$ equal columns $x_s[\eta] = q^{-2}\widehat{w}_s(\eta)$, $w_s = 2\cdot\mathbf{1}_{s(\mathrm{Im}\, d)^{-1}}$ (d is two-to-one onto a set of size $q/2$). By Parseval, $\sum_\eta \widehat{w}_s(\eta)^2 = 2q^2$ and $\widehat{w}_s(0) = q$, so $\|x_s\|^2 = q^{-2}$ on $\eta \neq 0$, and $\|M_{\mathcal{Z}}\|^2 \le \sum_{s\neq 0} n_s\|x_s\|^2 = (3^m - 2^m - q + 1)/q^2$. □

The function h enters only the phases of the $\mathcal{G}$ and $\mathcal{E}$ blocks; the $\mathcal{Z}$ block depends only on $\mathrm{Im}\, d$. The $\mathcal{G}$ estimate uses that p is a permutation and that d is affine; the $\mathcal{E}$ estimate uses APN in the direction ρ; oddness of m is not used.

6.3. **The exact $\mathcal{Z}$ block.**

Proposition 6.3 (Exact $\mathcal{Z}$ norm)**.** *For every $m \ge 3$ and all (p, d, h) under Assumption 2.3, $\|M_{\mathcal{Z}}\|^2 = (q/2-1)/q^2$. Consequently*

$$\tau^2 \;\le\; 4\Big(\frac{3}{4}\Big)^m - \frac{1}{2q} - \frac{1}{q^2}.$$

Proof. By Lemma 4.2 (which uses only (P2)), $\mathrm{Im}\, d = \{z : \mathrm{Tr}(\eta_0 z) = 1\}$ for a unique $\eta_0 \neq 0$. For $s \neq 0$ let $w_s(v) = 2\cdot[\mathrm{Tr}(\eta_0 s/v) = 1]$ for $v \neq 0$ and $w_s(0) = 0$, and $x_s(\nu) = q^{-2}\sum_v (-1)^{\langle \nu, v\rangle} w_s(v)$ for $\nu \neq 0$; by the proof of Theorem 6.2 these are the columns of the $\mathcal{Z}$ block, each repeated $n_s = 2^{m-\mathrm{wt}(s)} - 1$ times. One has $\sum_v w_s(v) = q$, $\sum_v w_s(v)^2 = 2q$, and for $s \neq t$, after the substitution $v \mapsto 1/v$, the conditions $\mathrm{Tr}(\eta_0 sv) = 1$ and $\mathrm{Tr}(\eta_0 tv) = 1$ are two independent affine conditions (the linear forms differ), so $\sum_v w_s(v)w_t(v) = 4\cdot q/4 = q$. By Parseval over all ν and removal of $\nu = 0$,

$$\langle x_s, x_t\rangle = q^{-4}\Big(q\sum_v w_s w_t - \big(\textstyle\sum_v w_s\big)\big(\sum_v w_t\big)\Big) = q^{-2}[s = t],$$

$$\sum_{\nu\neq 0} x_s(\nu) = q^{-2}\big(q w_s(0) - \textstyle\sum_v w_s(v)\big) = -\frac{1}{q}.$$

Hence on the extended space of all functions of $\nu \neq 0$, $M_{\mathcal{Z}}^* M_{\mathcal{Z}} = \sum_{s\neq 0} n_s x_s x_s^\top$ has the orthogonal eigenvectors x_s with eigenvalues n_s/q^2, the largest being $(q/2-1)/q^2$ ($\mathrm{wt}(s) = 1$); this gives $\|M_{\mathcal{Z}}\|^2 \le (q/2-1)/q^2$. The physical input space of the $\mathcal{Z}$ block is the space of functions on the $q-1$ points $(0, v)$, $v \neq 0$, with sum 0; extended by 0 at the origin, its image under the high-digit Walsh transform is exactly the set of coefficient vectors c with $c(0) = 0$ and $\sum_{\nu\neq 0} c(\nu) = 0$ (the value at the origin is $q^{-1/2}\sum_\nu c(\nu)$). For two distinct s, t of weight one ($m \ge 2$), $c = x_s - x_t$

satisfies both conditions and is an eigenvector for $(q/2-1)/q^2$. On these inputs the output low digit is $s = d(a)v \neq 0$, so the output fibre $s = 0$ (where the proof of Theorem 6.2 only uses a contraction) is not reached, and on the fibres $s \neq 0$ the projection onto $\perp$ is exactly the removal of $\xi = 0$. So the bound is attained by a physical input, and $\|M_{\mathcal{Z}}\|^2 = (q/2-1)/q^2$. The bound on τ^2 is Theorem 6.2 with the $\mathcal{Z}$ term replaced: $4(1.5^m-1)/q + 3/q + (q/2-1)/q^2 = 4(3/4)^m - 1/(2q) - 1/q^2$. $\square$

Since $\tau \geq \|M_{\mathcal{Z}}\| = ((q/2-1)/q^2)^{1/2}$, no bound $\tau = O(q^{-\gamma'})$ with $\gamma' > 1/2$ can hold uniformly; $1/2$ is the largest possible exponent.

6.4. **The $\mathcal{G}$ block: removal of h, and exponent $1/2$ for the inverse map with d of d_0 type.** For $u \notin \{0, \rho\}$, $k = K(u)$ and $a \in \mathbb{F}_q$ put $\mu_{u,a} = d(a)/k$, $y_{u,a} = D_u p(a)$, and for $z \neq 0$ let $\ell_{u,a}(z) = M^*_{\mu_{u,a}} z$. Let $g = q^2 \widehat{T}$ on the rows (u, η), $u \notin \{0, \rho\}$, $\eta \neq 0$, and the columns (s, ξ), $\xi \neq 0$; by Proposition 6.1 and the proof of Theorem 6.2, $\|M_{\mathcal{G}}\| = \|g\|/q^2$ (the $\mathcal{G}$ rows do not reach $s = 0$). Define

$$\mathsf{M}(p, d) = \max_{y, \nu \in \mathbb{F}_q} \#\big\{(u, a) : u \notin \{0, \rho\},\ D_u p(a) = y,\ d(a) = \nu K(u)\big\}.$$

Theorem 6.4 ($\mathcal{G}$ block, uniform in h)**.** *For every $m \geq 3$, all (p, d) under Assumption 2.3 and every h, $\|M_{\mathcal{G}}\|^2 \leq \mathsf{M}(p, d)/q$.*

Proof. Separation of h. In Proposition 6.1(1) write $C_a(s) = h(a) \oplus \mu_{u,a}(s \oplus E_{u,a})$ with $E_{u,a} = h(a \oplus u) \oplus h(a) \oplus d(a \oplus u)(a \star u)$. For a column with $s \star \xi = z$, $\langle z, C_a(s)\rangle = \langle z, h(a) \oplus \mu_{u,a} E_{u,a}\rangle \oplus \langle \ell_{u,a}(z), s\rangle$, so the sign of the atom (u, a) is $\varepsilon_{u,a}(z)\, \sigma^0_{u,a,z}(s, \xi)$ with $\varepsilon_{u,a}(z) = \pm 1$ independent of (s, ξ) and $\sigma^0_{u,a,z}(s, \xi) = (-1)^{\langle \xi, y_{u,a}\rangle + \langle s, \ell_{u,a}(z)\rangle}$ independent of h.

Factorization. Let Ψ be the 0/1 matrix with rows (u, η) and columns (u', a, z), $z \neq 0$, entry $[u = u'][\eta = M^*_{A_{u,a}} z]$, and Σ the matrix with rows (u, a, z) and columns (s, ξ), entry $[s \star \xi = z]\, \varepsilon_{u,a}(z) \sigma^0_{u,a,z}(s, \xi)$. Proposition 6.1(1) says $g = \Psi\Sigma$ (columns with $s \star \xi = 0$ vanish on both sides because $\eta \neq 0$). Since $A_{u,a} = d(a)d(a \oplus u)/k \neq 0$, $M^*_{A_{u,a}}$ is invertible, so for each row (u, η) and each a exactly one z occurs, and different rows have disjoint supports: $\Psi\Psi^\top = qI$, $\|\Psi\|^2 = q$. Every column (s, ξ) of Σ belongs to the single value $z = s \star \xi$, so $\Sigma\Sigma^\top$ is block diagonal over z, with blocks $\mathcal{B}_z = E_z \mathcal{B}^0_z E_z$, $E_z = \mathrm{diag}(\varepsilon_{u,a}(z))$, $\mathcal{B}^0_z = \Sigma^0_z \Sigma^{0\top}_z$. As E_z is a signed identity, $\|\mathcal{B}_z\| = \|\mathcal{B}^0_z\|$, and $\|g\|^2 \leq \|\Psi\|^2 \|\Sigma\|^2 = q \max_{z \neq 0} \|\mathcal{B}^0_z\|$; h has disappeared.

Frame bound. The non-zero spectrum of $\mathcal{B}^0_z$ equals that of the sum over all atoms (u, a) of the rank-one operators $\sigma^0_{u,a,z} \sigma^{0\top}_{u,a,z}$ on the columns $\{(s, \xi) : s \star \xi = z\}$. Each $\sigma^0_{u,a,z}$ is the restriction of a character of $\mathbb{F}_2^{2m}$, namely $\chi_P(\xi, s) = (-1)^{\langle \xi, y\rangle + \langle s, \ell\rangle}$ with $P = (y, \ell) = (y_{u,a}, \ell_{u,a}(z))$. If $n_z(P)$ atoms share the point P, then, in the semidefinite order,

$$\sum_{(u,a)} \sigma^0 \sigma^{0\top} = \sum_P n_z(P) \chi_P \chi_P^\top \preceq \big(\max_P n_z\big) \sum_{P \in \mathbb{F}_2^{2m}} \chi_P \chi_P^\top = \big(\max_P n_z\big)\, q^2 I$$

by orthogonality of characters. For $z \neq 0$ the map $\nu \mapsto M_\nu^* z$ is $\mathbb{F}_2$-linear and injective, so $\ell_{u,a}(z)$ determines $\mu_{u,a}$ and $\max_P n_z(P) = \mathsf{M}(p,d)$. Thus $\|\mathcal{B}_z^0\| \le \mathsf{M}q^2$, $\|g\|^2 \le \mathsf{M}q^3$ and $\|M_{\mathcal{G}}\|^2 = \|g\|^2/q^4 \le \mathsf{M}/q$. □

The absolute-value Schur estimate in the proof of Theorem 6.2 loses the factor $(3/2)^m$ in the cross terms between different u; the proof above avoids it by passing from the row Gram matrix to the atom Gram matrix, where h acts by a diagonal sign change. The theorem uses of the assumption only $d \neq 0$ and the definition of K.

Proposition 6.5 (Multiplicity for the inverse map)**.** *Let $p(x) = x^{q-2}$ and $K(x) = \sum_{j=0}^{e} k_j x^{2^j}$ with $k_e \neq 0$. Then $\mathsf{M}(p,d) \le 2^{e+1} + 2$. In particular $\mathsf{M} \le 10$ for $d = d_0$ and, more generally, for $d(x) = 1 \oplus K_0(x/\rho)$ with $K_0(x) = x^4 + x^2$ and any $\rho \neq 0$, m odd.*

Proof. Fix (y,ν). As p is a permutation, $y = D_u p(a)$ with $u \neq 0$ forces $y \neq 0$, and for given a at most one u works. For $a \notin \{0, 1/y\}$ the equation $1/(a+u) + 1/a = y$ gives $u = a^2 y/(1+ay)$. Multiplying $\varepsilon \oplus K(a) = \nu K(u)$ by $(1+ay)^{2^e} \neq 0$ gives a polynomial equation $\Phi(a) = 0$ of degree at most 2^{e+1} (the term $k_j(a^2y)^{2^j}(1+ay)^{2^e-2^j}$ has degree $2^e + 2^j$), and $\Phi(0) = \varepsilon \neq 0$ because $0 \in \operatorname{Im} K$ and $\varepsilon \notin \operatorname{Im} K$. Adding the two exceptional values of a gives the bound. For $d_0 = 1 \oplus K_0$, $e = 2$; the same holds for $K_0(x/\rho)$, whose kernel is $\{0, \rho\}$ and whose image is $\operatorname{Im} K_0 = \{\mathrm{Tr} = 0\} \not\ni 1$ for odd m. □

Corollary 6.6 (Exponent 1/2 for the inverse map with $d = 1 \oplus K_0(x/\rho)$)**.** *For $p(x) = x^{q-2}$, $m \ge 3$ odd, $d = 1 \oplus K_0(x/\rho)$ (any $\rho \neq 0$; $\rho = 1$ is d_0) and every h: $\tau^2 \le 13/q + (q/2-1)/q^2 < 13.5/q$, that is $\tau < 3.68\, q^{-1/2}$. For general PD data, $\tau^2 \le (\mathsf{M}(p,d)+3)/q + (q/2-1)/q^2$.*

Proof. $\tau^2 \le \|M_{\mathcal{G}}\|^2 + \|M_{\mathcal{E}}\|^2 + \|M_{\mathcal{Z}}\|^2$ (proof of Theorem 6.2), Theorem 6.4, the bound $\|M_{\mathcal{E}}\|^2 \le 3/q$, and Proposition 6.3. □

For general APN permutations p (or K of large 2-degree) a uniform bound on $\mathsf{M}(p,d)$ is not proved; in the computed cases ($p \in \{x^{-1}, x^3, x^5, x^{13}\}$, random K, $m \le 11$) $\mathsf{M} \le 11$. Uniformity of the exponent 1/2 over all PD data is therefore reduced to the h-free inequality $\max_{z \neq 0} \|\mathcal{B}_z^0\| \le Cq^2$ (Remark 6.7).

Remark 6.7 (What limits the exponent)**.** For general PD data, Theorem 6.2 gives the exponent $\gamma = 1 - \frac{1}{2}\log_2 3 \approx 0.2075$, and Theorem 6.4 improves it to 1/2 whenever $\mathsf{M}(p,d)$ is bounded. Numerically $\|M_{\mathcal{G}}\| \approx 2.2/q$ ($m = 3, 5, 7$), $\|M_{\mathcal{E}}\| \approx 0.8\, q^{-1/2}$ and $\|M_{\mathcal{Z}}\|^2 = (q/2-1)/q^2$ exactly. What remains open for exponent 1/2 uniformly over all PD data is the large-sieve type bound $\max_{z \neq 0} \|\mathcal{B}_z^0\| \le Cq^2$ with an absolute C; a bound $\sup \mathsf{M} < \infty$ suffices.

6.5. **Two words.** Let $M = \left[\begin{smallmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{smallmatrix}\right]$ have unit entries and unit determinant. The *alternating arrangement* A1 applies, in every step, B to each word with an independent uniform key in front, and then M. Its averaged

transition on two-word differences is $\mathcal{T}X = \mathrm{Perm}_M(T^\top XT)$, where X is a function on R^2 (an $N \times N$ array) and $(\mathrm{Perm}_M X)(M\beta) = X(\beta)$ (Proposition 2.6 applied word-wise; Section 2.6 is a different arrangement). Starting from $\pi_0 = e_\alpha$, $\alpha \neq (0,0)$, put $\pi_j = \mathcal{T}^j\pi_0$ and $D_j = \pi_j - U$, U the uniform distribution on $R^2 \setminus \{(0,0)\}$. The spaces P, Q, the quantities $\theta_2 = \|Q\mathcal{T}^2Q\|$, $\zeta_2 = \|Q\mathcal{T}^2P\|$, $\eta_{\mathrm{rss}} = \sqrt{2/q + 1/(q^2-1)}$ and the proof of the following reduction are in Appendix A (Proposition A.8): with $\nu^2 \le 2/(q-1) + \tau^2$,

$$\theta_2 \;\le\; \eta_{\mathrm{rss}} + \sqrt{\big(\nu^2 + (2+\sqrt{2})/\sqrt{q-1}\big)^2 + 2\nu^2}. \tag{2}$$

Corollary 6.8 (Unconditional two-word contraction)**.** *For all $m \ge 3$, all PD data and all h, (2) with Theorem 6.2 gives $\theta_2 = O(q^{-\gamma})$; with $x = (\sqrt{3}/2)^m$, $\theta_2 \le (1919/240)\,x < 8x$ for $m \ge 5$ (Lemma A.12).*

Theorem 6.9 (Step count with the proved exponent)**.** *Let $m \ge 5$, all PD data, all h, and A1 as above. With $x = (\sqrt{3}/2)^m = q^{-\gamma}$, for every integer $k \ge 0$,*

$$\|D_{2+2k}\|_2 \;\le\; \frac{8}{5}\,10^k\,q^{-1}x^{k-1}.$$

In particular $\|D_{14}\|_2 \le 1\,600\,000\,q^{-(1+5\gamma)} \le 1\,600\,000/N$, and for each word j,

$$\Pr_{\pi_{15}}[\beta_j = 0] \;\le\; \min\Big(1,\; \frac{11\,118\,751}{N}\Big).$$

Proof. Start. By Theorem 3.4 every entry of a non-zero row of T_B is at most $1/q$. Lemma 5.2 with $w = 2$, $T_i = U_j = T_B$, $b_\star = 1/q$, followed by the permutation Perm_M, gives $\|\pi_2\|_\infty \le 1/N$, hence $\|D_2\|_2 \le \|\pi_2\|_2 \le (\|\pi_2\|_\infty\|\pi_2\|_1)^{1/2} = 1/q$ ($D_2 \perp U$). For the cell projection, $\|PD\|_2^2 \le \sum_C m_C^2/|C| \le 1/(q^2-1)$, where m_C is the mass of π_2 in the cell C and every cell has at least q^2-1 points. Thus $a_2 := \|PD_2\| \le (q^2-1)^{-1/2}$ and $b_2 := \|QD_2\| \le 1/q$.

Comparison. By Lemma A.9, $(a_{j+2}, b_{j+2})^\top \le A_q(a_j, b_j)^\top$ entrywise with $A_q = \left[\begin{smallmatrix} \eta_{\mathrm{rss}}+q^{-4} & \eta_{\mathrm{rss}}(1+q^{-2}) \\ \zeta_2 & \theta_2 \end{smallmatrix}\right]$, $\zeta_2 \le 1$. Put $u = q^{-1/2}$. By Lemma A.12, $A_q \le \left[\begin{smallmatrix} 2u & 2u \\ 1 & 8x \end{smallmatrix}\right]$ and, with $\Delta = \mathrm{diag}(x, 1)$, the column sums of $\Delta^{-1}\left[\begin{smallmatrix} 2u & 2u \\ 1 & 8x \end{smallmatrix}\right]\Delta$ are $2u + x$ and $2u/x + 8x$, both at most $10x$ (since $u \le x^2$). For the weighted quantity $v_j = a_j/x + b_j$ (the ℓ^1 norm of $\Delta^{-1}(a_j, b_j)^\top$) this gives $v_{j+2} \le 10x\,v_j$.

Conclusion. $v_2 \le [(1-q^{-2})^{-1/2} + x]/(qx) \le 8/(5qx)$ because $(1-q^{-2})^{-1/2} \le 11/10$ and $x \le 1/2$. As $x \le 1$, $\|D_j\|_2 \le a_j + b_j \le v_j$, and $v_{2+2k} \le (10x)^k v_2$ gives the general bound. For $k = 6$, $x^5 \le 1/q$ because $3^5 < 2^8$; hence $\|D_{14}\| \le (8/5)10^6 q^{-2}$. Finally Lemma A.11 gives $|\Pr_{\pi_{15}}[\beta_j = 0] - 1/(N+1)| \le (1779/256)\|D_{14}\|_2$, and $1/(N+1) + (1779/256)\cdot 1\,600\,000/N \le 11\,118\,751/N$. □

The constants are far from sharp and the theorem is not a recommendation of a number of rounds for any small m; it is a statement, uniform

in h and in the PD data, that a fixed number of steps reaches the scale $1/N$. It concerns the key-averaged difference distribution of A1, not the $(2,2,1,1)$ arrangement of Section 5, and nothing here concerns a fixed key, a key schedule or PRP security.

For the inverse map with m odd and $d = 1 \oplus K_0(x/\rho)$, Corollary 6.6 supplies the hypothesis of the following proposition.

Proposition 6.10 (Step count under $\theta_2 \le Cq^{-1/2}$)**.** *Let $q \ge 32$, $u = q^{-1/2}$, and assume $\theta_2 \le Cu$. With A_q as above one has $0 \le A_q \le B(u) = \left[\begin{smallmatrix} 2u & 2u \\ 1 & Cu \end{smallmatrix}\right]$ entrywise, $\|B(u)^2\|_1 \le K_C u$ with $K_C = \max(C+24/5,\ 14/5+2C/5+C^2/5)$, hence $\|A_q^6\|_1 \le K_C^3 q^{-3/2}$, $\|D_{13}\|_2 \le \sqrt{2}K_C^3/N$, and the probability that one word is zero after the next step is at most $[1 + (1779/256)\sqrt{2}K_C^3]/N$.*

Proof. $\eta_{\mathrm{rss}} \le \frac{3}{2}u$ is equivalent to $q^2 - 4q - 1 \ge 0$; $q^{-4} \le u/2$; $\frac{3}{2}(1+q^{-2}) \le 2$; and $u \le 1/5$ for $q \ge 25$. The column sums of B^2 are $u(4u+4+C) \le (C+24/5)u$ and $u(2+(4+2C+C^2)u) \le (14/5+2C/5+C^2/5)u$. For entrywise non-negative matrices $0 \le A \le B$ implies $0 \le A^k \le B^k$, and the maximal column sum is monotone and submultiplicative. By Theorem 3.4 and Lemma A.10, $\|D_1\|_2 \le q^{-1/2}$ and $a_1 + b_1 \le \sqrt{2}\|D_1\|$; so $\|D_{1+2k}\|_2 \le \sqrt{2}q^{-1/2}\|A_q^k\|_1$, used with $k = 6$, and Lemma A.11 converts the ℓ^2 bound into a zero-word probability ($m \ge 5$). □

Corollary 6.11 (Inverse map with $d = 1 \oplus K_0(x/\rho)$: thirteen steps)**.** *For $p(x) = x^{q-2}$, odd $m \ge 5$, $d = 1 \oplus K_0(x/\rho)$ and every h, $\theta_2 \le (489/50)\, q^{-1/2}$; hence $\|D_{13}\|_2 \le 24\,405/N$ and $\Pr_{\pi_{14}}[\beta_j = 0] \le 169\,597/N$.*

Proof. Insert $\tau^2 \le 13/q + (q/2-1)/q^2$ (Corollary 6.6) and $\nu^2 \le 2/(q-1) + \tau^2$ in (2). Divided by u, each of the terms η_{rss}/u, $(\nu^2 + (2+\sqrt{2})/\sqrt{q-1})/u$ and ν/u is non-increasing in q for $q \ge 32$, so the ratio θ_2/u is at most its value at $m = 5$, which is below $489/50$ (outward rational evaluation). Then $K_C < 25.85$ for $C = 489/50$, and Proposition 6.10 gives $\sqrt{2}K_C^3 < 24\,405$ and $1 + (1779/256)\sqrt{2}K_C^3 < 169\,597$. □

7. Finite calibration, formalization and cost: the actual scope

The theorems of Sections 3–6 are proved on paper. This section records which parts were additionally checked on finite instances, which parts are checked by the Lean kernel, and what was measured about cost. A finite check is a calibration, not a proof of a general statement.

7.1. **Finite calibration.**

7.2. **Formalization.** The Lean developments use Lean 4.31.0-rc1 and Mathlib revision `d568c8c09630de097a046763c17b9ea99f95f950`; every listed declaration depends only on the standard axioms `propext`, `Classical.choice`, `Quot.sound`, with no `sorry`. The one-round bound $\varepsilon(B) \le 1/q$ of Theorem 3.4 is formalized for all directions on concrete carry coordinates. For $\mathrm{wt}(\rho) = 1$ and $w = 2$, the chain Proposition 3.5 →

Theorem 3.6 (one step) → Theorem 3.7(1) → Theorem 5.3(1) is checked in Lean on the real key averages on the additive group $(\mathbb{Z}/4)^m$ with the digit split φ_4, for the two-word mixer $(x, y) \mapsto (x + y,\, x + \Theta y)$ (`w2_Z4_theta_le`), and the mixer hypotheses are shown to be satisfiable ($m = 5$). The mixer is formalized as an additive automorphism, not as a matrix of ring scalars: in the pointwise ring $(\mathbb{Z}/4)^m$ a matrix with unit entries and unit determinant does not exist (`VacuousZ4`), so the earlier specialization with such scalars proves nothing about a concrete mixer and is superseded. That the additive group of $\mathrm{GR}(4, m)$ in a polynomial basis is $(\mathbb{Z}/4)^m$, and that multiplication by t is the companion matrix Θ of the lift $\hat{f}$, are used on paper; the local bound for $\mathrm{wt}(\rho) > 1$, general ω, general w, Theorem 3.9 and Section 6 are not formalized; of Section 4 the linear part on the actual definitions is formalized, the power sums are not. All Lean sources needed by these declarations are released as one source closure that builds from sources only.

Remark 7.1 (The mixer for general m)**.** Let $f \in \mathbb{F}_2[t]$ be irreducible of degree $m \geq 2$ and $\hat{f}$ its lift with coefficients in $\{0, 1\}$. Let Θ be the matrix of multiplication by t on $(\mathbb{Z}/4)[t]/(\hat{f}) \cong (\mathbb{Z}/4)^m$. Then $\det \Theta = \pm\hat{f}(0)$ and $\det(\Theta - I) = \pm\hat{f}(1)$ are odd, hence units of $\mathbb{Z}/4$, so Θ and $\Theta - I$ are invertible and $(x, y) \mapsto (x + y,\, x + \Theta y)$ is the mixer $\left[\begin{smallmatrix} 1 & 1 \\ 1 & t \end{smallmatrix}\right]$ of $\mathrm{GR}(4, m)$ in additive form, with partial maps I, I, I, Θ. This was checked exactly for all 69 binary irreducible f with $2 \leq \deg f \leq 8$ (inverse identities), and the agreement with multiplication by t in $\mathrm{GR}(4, m)$ on all pairs for $m = 3, 5$ (two polynomials each).

7.3. **Cost.** Counts are in a bit-level model with AND gates (XOR and NOT counted separately), for $m = 5$. They are reported to locate the constructions, not to claim an advantage. The Feistel network reaches the same constant with fewer AND gates. The differences that remain are structural (ring differences with carry versus XOR differences; $2m$-bit ring key additions; direction-wise structure; uniformity over all h; the integral structure of Section 4), not an ordering by cost. Ring-only and MPC costs were not measured.

TABLE 4. Finite checks (exact integer or rational arithmetic unless stated).

statement	instances	result
Theorem 3.4	$m = 5$: natural, $h = 0$, adversarial, three random h; $m = 7$: two p with natural, $h = 0$, random h; $m = 5$ with $\mathrm{wt}(\rho) = 3, 3, 5$	$\Delta(B) = q$ in every case; $\mathcal{Z}$-bound $2q/2^{\mathrm{wt}(s)}$ never violated
Theorem 3.7	$m = 5$, author DDT, exact matrix products	$N{\cdot}\mathrm{EDP}_2$: natural 185/128, $h = 0$ 311/128, adversarial 31/16, random h 303/256; x^{-1} with d_0 (ring-polynomial witness): 159/128; $\mathrm{wt}(\rho) = 3$ example 75/64 ($\leq 6 + 6/q$)
Theorem 3.9	$m = 5$: 11 tables (incl. $\mathrm{wt}(\rho) = 2, 3$); $m = 7$: 8 tables (incl. $\mathrm{wt}(\rho) = 4$)	identity of Lemma 3.13 holds for every $\mathcal{G}$ direction; $\mathcal{E}_{\mathcal{Z}} = 211/512$ ($m = 5$), 2059/8192 ($m = 7$); $\mathcal{E}_B \in [3.497, 3.559]$ ($m = 5$), $[3.345, 3.357]$ ($m = 7$)
Section 4 (d_0)	power sums $m = 3, 5, 7, 9$; coefficients: 40 cases each for $m = 3, 5$, 8 for $m = 7$; correction rule: all λ, 800 pairs; $m = 5$ all 1024 local keys for $h = a^3$	all agree; minimum number of non-zero c: 27 ($\tau = 0$, bound 18), 26 ($\tau = 1$, bound 19)
Lemma 4.2, Theorems 4.4–4.6, Proposition 4.7 (general d)	own arithmetic, $m = 3, 4, 5$; 19 maps d (random ρ, hyperplane and K, plus d_0); coefficients for 144/144/168 keys, all L and μ for $m = 3$ (45 056 keys); χ two choices; 550 affine h	all agree; minimal non-zero counts 6/13/28 ($\tau \neq \eta A$) and 7/12/28 ($\tau = \eta A$); Boolean degree of g_d exactly 2; pass rate in $\mathcal{Q}_2$ ($m = 3$, sample) 0.873–0.879
Proposition 6.1	author DDT $\to$ integer Walsh, $m = 3$ (24 576) and $m = 5$ (7 864 320 components), 16 tables, $\mathrm{wt}(\rho) = 1, 2, 3$	all agree; $\mathcal{G}$ entries in $\{0, \pm 4\}/q^2$ only

TABLE 5. Finite checks (continued).

statement	instances	result
Theorem 6.2	$m = 3, 5, 7$ (20 tables; numerical norms)	every block below its bound; $\|M_{\mathcal{Z}}\|^2 = (q/2-1)/q^2$ exactly
Proposition 6.3	independent code, $m = 3, 5$; d_0 and $1 \oplus K_0(x/\rho)$ ($\rho = 7, 11$); $h \in \{0, a^3, \text{random}\}$; 15 tables	integer Walsh of the $\mathcal{Z}$ rows $= x_s$; Gram $q^2[s = t]$ exactly; SVD norm and exact Rayleigh quotient of the witness $= (q/2-1)/q^2$
Theorem 6.4, Proposition 6.5	two independent implementations; $m = 3$: 16, $m = 5$: 21 tables; adversarial h (hill climbing); M for $p = x^{-1}$, $m = 3, \dots, 11$	$g = \Psi\Sigma$ entrywise; closed form of $\mathcal{B}_z^0$ exact; largest observed $\|M_{\mathcal{G}}\|^2 q/\mathsf{M}$ (floating point): 0.46287 ($m = 3$), 0.06390 ($m = 5$); $\mathsf{M} \le 5$ for d_0 type
Theorem 6.9, Corollary 6.11	every elementary inequality for $m = 5, \dots, 61$ (outward rational square roots); start lemma for $m = 3$, all 4095 two-word inputs, $M = [[1, 1], [1, t]]$	all hold; $\max \|\pi_2\|_\infty = 0.00464 < 1/64$ ($q = 8$ calibrates the start lemma only)
Example 4.13, Proposition 4.9	$m = 3$: all 64 carry-mask keys and all $512 \cdot 8$ linear (L, μ); $m = 5$: g_d for $\rho = 1$ and $\rho = 7$	$\ell(h) = 0$, no carry-mask key vanishes, $(L, \mu) = ((3, 0, 0), 0)$ vanishes (64 linear pairs in all); quadratic rank modulo affine $= 4 = m - 1$, the $(m-1)$-AND circuit agrees at all points
Theorem 5.7	decomposition: toy $m = 2$, $w = 2$, all 16 777 216 candidates, 32 cases; $m = 3$ sample of 58 035	accepted sets identical
Theorem 5.7(3)	$M = 5$, $Q = 3$, all 120 permutations	RP law = RF | D

TABLE 6. What is checked in Lean, and what is an input.

file	checked content	not covered / input
`WarmStart`	arithmetic of the warm-started recursion: from initial values $\le x$ the bound $4x^2$; $\le 4/q^2$ for real $q \ge 32$	the counts of Section 3 on a concrete ring
`SuffixReset`, `ResetCompose`	DDT row/column sums on any finite abelian group; reset identity of Lemma 5.1; two applications (four layers)	the bound on the first part is an input
`W2Bridge`	averaged kernel of a parallel layer with product keys is the tensor product; real averaged DP of the $w = 2$ superbox equals the abstract MDS kernel; $(2, 2, 1, 1)$, $w = 2$: $\mathbb{E}\mathrm{DP} \le b_\star^2$	the one-word bound $b_\star$ (Theorem 3.7) is an input; general w; identification with the reference code
`CarryPD`	two-digit addition with an *arbitrary* carry function, $B = J \circ F$, real DDT: in $\mathcal{G}$ directions ($d(a+u) = d(a) + k$, $k \ne 0$) $\delta_B \le q$ for all p, h, β	$\mathcal{Z}$ and $\mathcal{E}$ directions; Theorem 3.7; connection to `W2Bridge`

TABLE 7. What is checked in Lean (continued).

file	checked content	not covered / input
`CarryZ`, `CarryE_v2`	concrete carry $\star$ (product in a fixed basis $e : \mathbb{F}_q \simeq \mathbb{F}_2^n$); fibres of $x \mapsto x \star s$ have $\le q/2$ points for $s \neq 0$; $\mathcal{Z}$ direction $\delta_B \le q$ ($d \neq 0$, d two-to-one); $\mathcal{E}$ direction $\delta_B \le q$ (p APN along ρ)	exact fibre size $q/2^{\mathrm{wt}\,s}$
`OneRound`	all $\alpha \neq 0$: $\delta_B \le q$ and DP $\le 1/q$, for $d = \varepsilon \oplus K$, $\ker K = \{0, \rho\}$, $d \neq 0$, p APN, every h (two-to-one derived; the $\mathcal{G}$ lemma is that of `CarryPD`)	—
`Transport_v2`	for any group G and bijection $\varphi : G \to \mathbb{F}_q^2$ carrying $+$ to the carry addition: ddt = ddtC exactly; keyed families of PD layers have avgDP $\le 1/q$ in the definitions of `W2Bridge`	the identification $\mathrm{GR}(4, m) \cong (\mathbb{F}_q^2, \oplus_\star)$ (Lemma 2.1) is the hypothesis φ; the two-block bound $4/N$
`RepairCore_v2`	$A^2 + B \neq 0 \Rightarrow$ top coefficients not both 0; root count $\ge q - (s-1)$; for any finite $\mathbb{F}_q$-module V and $\ell(g) = 1$: surjectivity, $\ell(h + \lambda g) = \ell(h) + \lambda$, exactly one failing λ, $q \cdot \#\ker \ell = \lvert V \rvert$	$\ell = \ell_d$, power sums, Lucas, $\ell_d(g_d) = 1$, $\Psi = \Phi^2$
`ColsB_v3`, `ColsF_v2`	the rows of Proposition 3.5 as counts of the real DDT of B and of F on concrete carry coordinates, for $\mathrm{wt}(\rho) = 1$: $\mathcal{Z} \to \mathcal{Z}$ 0, $\mathcal{Z} \to \mathcal{E}$ column $\le q^2/2$, $\mathcal{E} \to \mathcal{Z}$, $\mathcal{E} \to \mathcal{E}$ columns $\le 2q$, $\mathcal{G} \to \mathcal{Z} \le 2$, $\mathcal{G} \to \mathcal{E} \le 4$; final layer: low digit 0 only from $\mathcal{Z}$, $\mathcal{Z}$ column $\le q^2$, $\mathcal{E}$ column $\le 2q$, $\mathcal{G}$ entries ≤ 2	general ω (only $\le 1/2$ for $\mathcal{Z} \to \mathcal{E}$), so Theorem 3.7(2)
`TwoStep`, `Bridge`	Theorem 3.6 one step from the warm start and the final layer: $(T_B^2 T_F)(\alpha, \gamma) \le 4/N$ ($q \ge 12$); the real key average of S_2 (with whitening) equals $T_B^2 T_F$; hence `hb1`–`hb4` of `W2Bridge` are theorems and the $(2, 2, 1, 1)$, $w = 2$ average is $\le (4/N)^2$	$\mathrm{wt}(\rho) = 1$ only; $t \ge 2$ of Theorem 3.6; the identification φ of $\mathrm{GR}(4, m)$ with $(\mathbb{F}_q^2, \oplus_\star)$ remains a hypothesis
`RepairLink_v2`	ℓ_d on actual definitions (A_d, B_d over a transversal, Frobenius) is $\mathbb{F}_q$-linear; $\Delta_{g_d} = r$; repair, one failure and pass count $1 - 1/q$ for the actual ℓ_d	$\ell_d(g_d) = 1$ (power sums p_2, p_3) is the only hypothesis; that the index set is a transversal is not stated in Lean

TABLE 8. What is checked in Lean (continued): the concrete two-word mixer.

file	checked content	not covered / input
`DigitSplit_v2`	the digit split $\varphi_4 : (\mathbb{Z}/4)^m \to \mathbb{F}_q^2$, $x = a + 2b$ coordinatewise, carries $+$ to the carry addition (`phi4_add`)	the ring multiplication of $\mathrm{GR}(4, m)$ is not used; a statement there that takes the mixer entries in the *pointwise* ring $(\mathbb{Z}/4)^m$ has no instance (`VacuousZ4`)
`VacuousZ4`	in the pointwise ring $(\mathbb{Z}/4)^m$, $m \geq 1$, four unit entries and a unit determinant are incompatible (`w2_Z4_mixer_hyps_false`); hence the earlier specialization of the two-word bound to that ring is vacuous and is not counted as a verification	—
`MixAdd`	two-word MDS kernel lemma over any finite abelian group A for an additive automorphism M of A^2 whose four partial maps are bijective (`mds2A_le`); real key average of the $w = 2$ superbox $=$ this kernel; $(2, 2, 1, 1)$, $w = 2$: $\mathbb{E}\mathrm{DP} \leq \varepsilon^2$ (`w2A_split_tail_le`)	the one-word bound ε is an input
`ThetaZ4_v3`	mixer $M(x, y) = (x + y,\, x + \Theta y)$ on $A = (\mathbb{Z}/4)^m$ for an additive automorphism Θ with $\Theta - I$ invertible; with the actual local blocks S_2 and φ_4: $(2, 2, 1, 1)$, $w = 2$ average $\leq (4/N)^2$ (`w2_Z4_theta_le`); explicit witness $m = 5$, $\Theta =$ multiplication by t in $(\mathbb{Z}/4)[t]/(t^5 + t^2 + 1)$, with Θ^{-1} and $(\Theta - I)^{-1}$ (`theta5`, `theta5_mixer_mds`, `w2_Z4_theta5_le`)	PD data (p, K, ε) with $\mathrm{wt}(\rho) = 1$ are hypotheses; for general m the choice of Θ is on paper (Remark below)

TABLE 9. Bit-model counts ($m = 5$).

object	AND	averaged differential bound
three-round Feistel with x^3, independent XOR keys	75	$4/N$ ([4], Thm. 2; bijective round function, ≥ 3 rounds)
S_2 with the natural table of [2]	145	$4/N$ (Theorem 3.7)
S_2 with an inverse-based PD table	235	$4/N$ (Theorem 3.7)
one F, PD table with $h = a^3$	75 (98 XOR, 1 NOT)	—
correction g_d alone (Proposition 4.9)	$m-1$, minimal (4 for $m = 5$)	— (cost of the correction only, not of h)

8. Limits and remaining design questions

8.1. **What the theorems cover.** Each theorem fixes its quantifiers as follows. Theorems 3.4 and 3.9: all h, all ρ, all (p, d) under Assumption 2.3 (for Theorem 3.9 the permutation property of p is not used), exact counts. Theorem 3.7: odd $m \geq 5$, all h; constant 4 only for $\mathrm{wt}(\rho) = 1$ in the fixed basis, $(6 + 6/q)$ otherwise; average over independent uniform keys and whitening. Theorem 5.3: the $(2, 2, 1, 1)$ arrangement with three mixings and reset keys L_2, L_3, M_1 MDS, M_2, M_3 invertible, average over $10w$ independent uniform key words. Theorem 5.7: non-adaptive queries on pre-fixed disjoint slices; the real construction is accepted for every key; false acceptance bounded by a union over N^{3w} candidates. Section 4: $p(x) = x^{q-2}$, every d satisfying (P2), every $m \geq 3$ for the algebra (odd $m \geq 5$ when combined with Section 3), every local key; the certificate is sufficient for every $\mathbb{F}_2$-linear L, and its necessity is not asserted (Example 4.13); the degree statement concerns h only; Proposition 4.9 concerns g_d alone. Theorem 6.2, Proposition 6.3, Theorem 6.4: every $m \geq 3$, all PD data, all h; averaged transitions. Corollary 6.6, Corollary 6.11: $p(x) = x^{q-2}$, odd m, $d = 1 \oplus K_0(x/\rho)$, all h. Theorem 6.9: $m \geq 5$, all PD data, all h, the alternating arrangement A1 with independent uniform keys in every step and a two-word mixing with unit entries and determinant.

8.2. **What remains open or is not addressed.**

- The one-word contraction has exponent $\gamma \approx 0.2075$ for all PD data (Theorem 6.2) and the optimal exponent $1/2$ for the inverse map with m odd and $d = 1 \oplus K_0(x/\rho)$ (Corollary 6.6). For the inverse map with another admissible $d = \varepsilon \oplus K$, K of linearized degree e, the constant becomes $2^{e+1} + 5$ (Proposition 6.5), which is not uniform in m when e grows with m. Exponent $1/2$ uniformly over all PD data is reduced to the h-free bound $\max_z \|\mathcal{B}_z^0\| \leq Cq^2$ and is open. The step counts (Theorem 6.9, Corollary 6.11) have large constants; they are statements about the scale $1/N$, not round recommendations. Numerical contraction constants are estimates.
- No fixed-key differential bound, no linear or algebraic bounds, no related-key statement.
- No key schedule; the $10w$ independent key words are the randomness of the averaging model.
- No lower bound on attack time; the implemented enumerations are upper bounds for one method. Meet-in-the-middle and character-sum (Fourier) decompositions of the candidate search, higher-order integrals and division-property analyses were not carried out.
- The local correction removes one family of local zero sums inside an S_2 block; it does not remove the word-level zero sum used by Theorem 5.7, and it is not a statement about integral resistance of the arrangement.

- The classification of h with $\ell_d(h) = 0$ is not attempted; the vanishing example of Example 4.12 is a single instance. Non-inverse p and non-affine d are not treated in Section 4.
- Cost: no claim of minimal AND count or of an advantage over known constructions; ring-only and MPC costs are unmeasured.

8.3. **Design questions left for later work.** The arrangement $(2, 2, 1, 1)$ and the number of mixing layers are fixed in this paper; we do not add layers to raise attack cost. A design that addresses the word-level integral of Lemma 5.6, a key schedule, and fixed-key statements would be required before any construction built from these components could be considered as a cipher.

Data and code availability

The finite checks of Section 7 and the Lean 4 sources with their build records are provided in the public repository `https://github.com/hirokifukui/residue-apn-carry`, archived at Zenodo, `https://doi.org/10.5281/zenodo.23330906` (all versions). The directory `checks/` contains three independently written checkers, with their expected outputs, for Proposition 6.3, Theorem 6.9 with Corollary 6.11, and Example 4.13 with Proposition 4.9 (including the gate count of the correction g_d in Table 9). For every other row of Tables 4, 5 and 9, and for the check reported in Remark 7.1, the directory `evidence/` contains the unedited scripts that produced the reported numbers, the logs, exit codes and result files of those runs, and the input tables (included, or regenerated by included scripts); the index `EVIDENCE_COVERAGE.tsv` lists, row by row, the script, the recorded output and whether the row is re-run. The script `run_all_representatives.sh` re-runs one representative command per row (Python 3 with NumPy and SymPy; about ten minutes; for the gate count of one F with $h = a^3$ through a small added wrapper that leaves out a cross-check against a table that is not distributed; for the adversarial search of Theorem 6.4 with $m = 3$ by recomputing the 40 recorded witnesses, re-running the search itself being optional) and compares the result with the recorded output. For the following entries the numbers rest on the recorded outputs only, because the runs are long: the $m = 7$ instances of Theorems 3.4 and 6.2, the $m = 5$ instances of Theorem 6.4 and Proposition 6.5, and 29 of the 32 toy cases and the $m = 3$ sample of Theorem 5.7. A finite check is a calibration, not a proof.

Use of generative AI

Generative AI systems, including Anthropic Claude and OpenAI GPT models, were used throughout this project to assist with the exploration and comparison of proof strategies and derivations, adversarial review of successive drafts, the design and implementation of the checker scripts for the finite computations, tactic-level search and code drafting for the Lean

formalization, and drafting and revision of the manuscript. The author made all final mathematical and editorial decisions, checked every definition, statement, proof, and citation included in the manuscript, and assumes sole responsibility for the entire content. As described in Section 7, designated finite computations and selected proof components were additionally checked by checker scripts or by the Lean kernel; these checks do not rely on accepting AI-generated mathematical assertions, and a finite check is a calibration, not a proof.

Appendix A. The two-word reduction

Throughout, Assumption 2.3 holds, $T = T_B$, $W = R \setminus \{0\}$, and M is a 2×2 matrix over R whose entries and determinant are units. All spaces carry the counting-measure inner product.

A.1. **Spaces and operators.** Let $V_0 = \{X : R^2 \to \mathbb{R},\ X(0,0) = 0\}$, $U = \mathbf{1}_{R^2\setminus\{(0,0)\}}/(N^2-1)$, and work in $V_0 \ominus U$. For $X \in V_0$ put $SX = T^\top XT$ (that is, $(SX)(\gamma_1,\gamma_2) = \sum_\beta T(\beta_1,\gamma_1)X(\beta_1,\beta_2)T(\beta_2,\gamma_2)$) and $\mathcal{T}X = \mathrm{Perm}_M(SX)$ with $(\mathrm{Perm}_M X)(M\beta) = X(\beta)$. Since $T^\top$ is doubly stochastic with $T(0,\cdot) = e_0$ and $T(\alpha,0) = 0$ for $\alpha \neq 0$, and Perm_M is an orthogonal permutation fixing $(0,0)$, $\mathcal{T}$ maps V_0 to itself, $\mathcal{T}U = U$, $\|\mathcal{T}\| \le 1$, and $\mathcal{T}$ preserves $V_0 \ominus U$.

Cells. For $s,t \in \mathbb{F}_q$ let $f_s = \{\beta \in R : \mathrm{lo}\,\beta = s\}$ (q points). The cells of $R^2 \setminus \{(0,0)\}$ are $f_s \times f_t$ for $(s,t) \neq (0,0)$ and $(f_0 \times f_0) \setminus \{(0,0)\}$; every cell has at least $q^2 - 1$ points. $\Lambda' \subset V_0$ is the space of functions constant on cells, $P = \Pi_{\Lambda'}$ and $Q = I - P$ (on $V_0 \ominus U$; note $U \in \Lambda'$). In one word, λ is the space of functions on R constant on the fibres f_s, $\tilde{\lambda}$ the functions on W constant on $f_s \cap W$, and $\perp \subset \ell^2(W)$ the functions with sum 0 on every $f_s \cap W$; $T' = T|_{W\times W}$ and $\tau = \|\Pi_\perp T'^\top \Pi_\perp\|$ as in Theorem 6.2.

A.2. **The slow mode.**

Lemma A.1 (L1)**.** Perm_M *maps cells to cells and fixes* $(0,0)$*; hence* $\mathrm{Perm}_M P = P\,\mathrm{Perm}_M$.

Proof. Reduction modulo $2R$ is a ring homomorphism, so $\mathrm{lo}(M\beta) = \bar{M}\,\mathrm{lo}(\beta)$ with $\bar{M}$ invertible over $\mathbb{F}_q$. □

Lemma A.2 (L2)**.** *In one word,* $\Pi_\lambda T^\top \Pi_\lambda$ *is the projection onto the constants: the low digit of the output, averaged over an input fibre* f_u*, is uniform.*

Proof. Use Lemma 3.2. $\mathcal{G}$ ($u \notin \{0,\rho\}$): for each (a,s) exactly one b gives low difference s, so every row of the fibre has a uniform low marginal. $\mathcal{Z}$ ($u = 0$): $v = 0$ is the zero difference (output 0), and for $v \neq 0$ the low output $s = d(a)v$ is uniform on $v\,\mathrm{Im}\,d$ (each value with probability $2/q$); averaging over v gives $q^{-1}([s=0] + [s\neq 0]\cdot(2/q)\cdot\#\{v \neq 0 : s \in v\mathrm{Im}\,d\}) = q^{-1}([s = 0] + [s \neq 0]) = 1/q$. $\mathcal{E}$ ($u = \rho$): the low output $s_v(a)$ is a bijective function of v for each a, so the average over v is uniform. □

Lemma A.3 (L3)**.** $\|\Pi_\lambda T^\top \Pi_{\lambda^\perp}\| \le \sqrt{2/q}$.

Proof. For $Y \in \lambda^\perp$ (sum 0 on every input fibre) the low marginal of $T^\top Y$ is $m(s) = \sum_\alpha Y(\alpha)K(\alpha, s)$ with $K(\alpha, s) = \Pr[\text{lo of output} = s \mid \alpha]$, and $\|\Pi_\lambda T^\top Y\|^2 = q^{-1}\sum_s m(s)^2$. Rows in $\mathcal{G}$ have constant K (Lemma A.2), so their contribution vanishes. On the $\mathcal{Z}$ fibre, $K_Z K_Z^\top = \mathrm{diag}(1) \oplus q^{-1}(I + J)$, because $|v\mathrm{Im}\, d \cap v'\mathrm{Im}\, d| = q/4$ for $v \ne v'$ ($\mathrm{Im}\, d$ is an affine hyperplane); on vectors of sum 0 its largest Rayleigh quotient is 1, so $\|K_Z^\top Y_Z\| \le \|Y_Z\|$. On the $\mathcal{E}$ fibre the $q \times q$ matrix K_E is doubly stochastic, so $\|K_E^\top Y_E\| \le \|Y_E\|$. Hence $\|m\| \le \|Y_Z\| + \|Y_E\| \le \sqrt{2}\|Y\|$ and $\|\Pi_\lambda T^\top Y\| = q^{-1/2}\|m\| \le \sqrt{2/q}\,\|Y\|$. □

Lemma A.4 (L4)**.** *On $V_0 \ominus U$, $P\mathcal{T}P = -q^{-2}\psi\psi^\top/\|\psi\|^2$ with $\psi = -\mathbf{1}_c + q^{-2}\mathbf{1}_{R'}$, where $c = (f_0 \times f_0) \setminus \{(0,0)\}$ and R' is the set of the other non-zero states; in particular $\|P\mathcal{T}P\| = 1/N$.*

Proof. For a cell $\sigma = (s,t) \ne (0,0)$, $S\mathbf{1}_{f_s \times f_t} = (T^\top \mathbf{1}_{f_s}) \otimes (T^\top \mathbf{1}_{f_t})$; by Lemma A.2 each factor has mass 1 in every fibre and value $[s = 0]$ at 0. Hence the cell projection of $S\mathbf{1}_\sigma$ equals q^{-2} on R' and $1/(q^2-1)$ on c, while $S(\mathbf{1}_{f_0 \times f_0} - e_{00})$ projects to q^{-2} on R' and 0 on c. For $X = \sum_\sigma x_\sigma \mathbf{1}_\sigma \in \Lambda'$ of mass 0, $q^2 \sum_{\sigma \ne 00} x_\sigma = -(q^2-1)x_{00}$, and substituting gives $PSX = (x_{00}/q^2)\psi$; Perm_M preserves c and R' (Lemma A.1). Finally $\langle \psi, X\rangle/\|\psi\|^2 = -x_{00}$. □

Lemma A.5 (L5)**.** $\eta := \|P\mathcal{T}Q\| \le \eta_{\mathrm{rss}} = \sqrt{2/q + 1/(q^2-1)}$.

Proof. Work in $\ell^2(R^2) \supset V_0$. There, the cell-constant functions together with e_{00} form $(\lambda \otimes \lambda) \oplus \mathrm{span}\{e'\}$, $e' = e_{00} - q^{-2}\mathbf{1}_{f_0 \times f_0}$, an orthogonal sum with $\|e'\|^2 = 1 - q^{-2}$; for $Y \in V_0$ the projection onto Λ' is the projection onto this space, because $Y \perp e_{00}$. By Lemma A.1 Perm_M may be removed. For $X \in \Lambda'^\perp \subset (\lambda \otimes \lambda)^\perp$ split X into its $\lambda \otimes \lambda^\perp$, $\lambda^\perp \otimes \lambda$ and $\lambda^\perp \otimes \lambda^\perp$ parts; $\Pi_{\lambda\otimes\lambda} S = (\Pi_\lambda T^\top) \otimes (\Pi_\lambda T^\top)$ and Lemma A.3 bound the three images, which are mutually orthogonal, by $\sqrt{2/q}$ times the norms, so the $\lambda \otimes \lambda$ component is at most $\sqrt{2/q}\|X\|$. The e' component is $|\langle (T \otimes T)e', X\rangle|/\|e'\| = q^{-2}|\langle (T\mathbf{1}_{f_0}) \otimes (T\mathbf{1}_{f_0}), X\rangle|/\|e'\| \le q^{-2} \cdot q\|X\|/\sqrt{1-q^{-2}}$ (the vector $(T\mathbf{1}_{f_0}) \otimes (T\mathbf{1}_{f_0})$ has norm at most q). The two components are orthogonal. □

A.3. **One word, and the mixed Gram estimates.**

Lemma A.6 (W1–W4)**.** *(W2) $\ell_W := \|\Pi_{\tilde\lambda} T'^\top \Pi_\perp\| \le \sqrt{2/(q-1)}$. (W3) $\nu := \|T'^\top \Pi_\perp\|$ satisfies $\nu^2 \le 2/(q-1) + \tau^2$; hence, with $\mathcal{K} = \perp \otimes \perp$ and $\mathcal{R} = (\tilde\lambda \otimes \perp) \oplus (\perp \otimes \tilde\lambda)$, $\|S\Pi_\mathcal{K}\| \le \nu^2$ and $\|S\Pi_{\mathcal{R}\oplus\mathcal{K}}\| \le \sqrt{2}\nu$. (W4) S preserves the axis $\mathrm{Ax} = e_0 \otimes \ell^2(W) \oplus \ell^2(W) \otimes e_0$ and the interior $\ell^2(W \times W)$, and $\|S|_{\mathrm{Ax}}\| \le 1$. (W1, not used below) $\|\Pi_{\tilde\lambda \ominus 1} T'^\top \Pi_{\tilde\lambda \ominus 1}\| = 1/q$.*

Proof. (W2) Extend $Y \in \perp$ by 0 at the origin; it lies in $\lambda^\perp$ on R and $(T^\top Y)(0) = Y(0) = 0$. Averaging over $q-1$ instead of q points in f_0

multiplies the bound of Lemma A.3 by at most $\sqrt{q/(q-1)}$. (W3) $T'^{\top}$ preserves mass, so $T'^{\top}Y$ has no constant component for $Y \in \perp$; split it into its $\tilde{\lambda}$ and $\perp$ parts. On the interior $S = T'^{\top} \otimes T'^{\top}$, so $\|S\Pi_{\mathcal{K}}\| = \nu^2$; writing $\mathcal{R} \oplus \mathcal{K} = (\ell^2(W) \otimes \perp) \oplus (\perp \otimes \tilde{\lambda})$ (orthogonal inputs, each image of norm at most ν) gives $\sqrt{2}\nu$. (W4) $T^{\top} e_0 = e_0$ and $T^{\top}\ell^2(W) \subset \ell^2(W)$. (W1) is a rank-one computation from Lemma A.2 and is recorded only for completeness. □

Lemma A.7 (G0, G1)**.** *(G0) If $X \perp \Lambda'$, then $\|\Pi_{\tilde{\lambda}\otimes\tilde{\lambda}} X_{\mathrm{int}}\| \le \sqrt{2/(q-1)}\,\|X_{\mathrm{Ax}}\|$. (G1) $\|(I - \Pi_{\mathcal{K}})\,\mathrm{Perm}_M\,\Pi_{\mathrm{Ax}}\| \le 2/\sqrt{q-1}$.*

Proof. (G0) On cells (s,t) with $s,t \neq 0$ the sum of X is 0 and there are no axis points, so the $\tilde{\lambda}\otimes\tilde{\lambda}$ component vanishes. A cell $(0,t)$, $t \neq 0$, has q axis points and $q(q-1)$ interior points; the interior sum is minus the axis sum, which is at most $\sqrt{q}$ times the axis norm, so the component has norm at most $\|X_{\mathrm{Ax}}|_{\mathrm{cell}}\|/\sqrt{q-1}$. The cell c contributes in the same way with $2(q-1)$ axis points. (G1) $\mathrm{Perm}_M(e_0 \otimes g)$ is supported on the line $\{(m_{12}r, m_{22}r)\}$ (both coefficients units), which lies in the interior; for fixed second coordinate at most one first coordinate in a fibre lies on the line, and conversely. Hence each of the two one-word $\tilde{\lambda}$ projections of the line function has norm at most $\|g\|/\sqrt{q-1}$; as $I - \Pi_{\mathcal{K}} = \Pi_{\tilde{\lambda}} \otimes I + \Pi_{\perp} \otimes \Pi_{\tilde{\lambda}}$ is an orthogonal sum, the non-$\mathcal{K}$ part is at most $\sqrt{2/(q-1)}\|g\|$. The two halves of the axis are orthogonal inputs, which gives $\sqrt{2}\cdot\sqrt{2/(q-1)} = 2/\sqrt{q-1}$. □

Proposition A.8 (D2)**.**

$$\theta_2 = \|Q\mathcal{T}^2 Q\| \le \eta_{\mathrm{rss}} + \sqrt{\left(\nu^2 + (2+\sqrt{2})/\sqrt{q-1}\right)^2 + 2\nu^2}.$$

Proof. With $A = P\mathcal{T}P$, $B = P\mathcal{T}Q$, $C = Q\mathcal{T}P$, $D = Q\mathcal{T}Q$ one has $Q\mathcal{T}^2Q = CB + D^2$, $\|C\| \le 1$ and $\|B\| \le \eta_{\mathrm{rss}}$ (Lemma A.5). Let $X = QX$, $a = \|X_{\mathrm{Ax}}\|$, $b = \|X_{\mathrm{int}}\|$. Then $SX = SX_{\mathrm{Ax}} + S\Pi_{\tilde{\lambda}\otimes\tilde{\lambda}}X_{\mathrm{int}} + S\Pi_{\mathcal{R}\oplus\mathcal{K}}X_{\mathrm{int}}$, the last two terms of norm at most $\sqrt{2/(q-1)}\,a$ (Lemma A.7, G0) and $\sqrt{2}\nu b$ (Lemma A.6). Write $Y = \mathrm{Perm}_M SX = L + E$ with $L = \mathrm{Perm}_M SX_{\mathrm{Ax}}$, $\|L\| \le a$, and $\|E\| \le \sqrt{2/(q-1)}\,a + \sqrt{2}\nu b$. Since $\mathcal{K} \perp \Lambda'$, $QY = \Pi_{\mathcal{K}}L + Q(I - \Pi_{\mathcal{K}})L + QE$, and $D^2X = Q\,\mathrm{Perm}_M S\,QY$ with $\|S\Pi_{\mathcal{K}}L\| \le \nu^2 a$ and $\|(I - \Pi_{\mathcal{K}})L\| \le (2/\sqrt{q-1})a$ (G1). Collecting, $\|D^2X\| \le (\nu^2 + (2+\sqrt{2})/\sqrt{q-1})a + \sqrt{2}\nu b$, and Cauchy–Schwarz with $a^2 + b^2 = \|X\|^2$ gives the bound for $\|D^2\|$. □

A.4. **Comparison, start and the line lemma.**

Lemma A.9 (comparison)**.** *For $a_j = \|PD_j\|$, $b_j = \|QD_j\|$: $a_{j+1} \le q^{-2}a_j + \eta b_j$, $b_{j+1} \le a_j + b_j$, hence $a_{j+2} \le (\eta + q^{-4})a_j + \eta(1+q^{-2})b_j$ and $b_{j+2} \le \zeta_2 a_j + \theta_2 b_j$ with $\zeta_2 = \|Q\mathcal{T}^2P\| \le 1$.*

Proof. $PD_{j+1} = P\mathcal{T}PD_j + P\mathcal{T}QD_j$ with Lemmas A.4 and A.5; $\|QD_{j+1}\| \le \|D_j\| \le a_j + b_j$; iterate once for a_{j+2}, and split $D_j = PD_j + QD_j$ under $Q\mathcal{T}^2$ for b_{j+2}. □

Lemma A.10 (start)**.** $\|D_1\|_2 \le \|\pi_1\|_2 \le q^{-1/2}$ *and* $\|\pi_2\|_\infty \le 1/N$.

Proof. $\|\pi_1\|_2^2 = \mathcal{E}(\alpha_1)\mathcal{E}(\alpha_2)$ with the row energies $\mathcal{E}(\alpha) = \sum_\beta T(\alpha,\beta)^2$, which equal 1 for $\alpha = 0$ and are at most $\max_\beta T(\alpha,\beta) \le 1/q$ otherwise (Theorem 3.4); at least one word of α is non-zero. The second statement is Lemma 5.2 ($w = 2$, $b_\star = 1/q$) followed by the permutation Perm_M. □

Lemma A.11 (line lemma)**.** *For $m \ge 5$ and every $j \in \{1,2\}$, $|\Pr_{\pi_{r+1}}[\beta_j = 0] - 1/(N+1)| \le (1779/256)\|D_r\|_2$.*

Proof. Let $A_j = \{\beta \ne (0,0) : \beta_j = 0\}$ and $f = \mathcal{T}^*\mathbf{1}_{A_j}$, so

$$f(\alpha) = \sum_\delta T(\alpha_1,\delta_1)T(\alpha_2,\delta_2)[(M\delta)_j = 0].$$

The condition $(M\delta)_j = 0$ is the graph $\delta_2 = \phi(\delta_1)$ of a bijection (m_{j2} is a unit), so by Cauchy–Schwarz $f(\alpha) \le (\mathcal{E}(\alpha_1)\mathcal{E}(\alpha_2))^{1/2}$ and $\|f\|_2 \le \mathcal{E}_B \le 1779/256$ (Theorem 3.9). As $\mathcal{T}U = U$ and $D_r \perp U$, $\Pr_{\pi_{r+1}}[\beta_j = 0] = \langle f, U\rangle + \langle f, D_r\rangle$ with $\langle f, U\rangle = |A_j|/(N^2-1) = 1/(N+1)$. □

Lemma A.12 (arithmetic)**.** *Let $m \ge 5$, $x = (\sqrt{3}/2)^m$, $u = q^{-1/2}$. Then $x \le 1/2$, $u/x \le 3/8$, $u \le x^2$, $\eta_{\mathrm{rss}} \le \frac{3}{2}u$, $2/(q-1) \le x^2/2$, $(2+\sqrt{2})/\sqrt{q-1} \le \frac{18}{5}u$; with Theorem 6.2, $\nu^2 \le \frac{11}{2}x^2$ and $\theta_2 \le \frac{1919}{240}x < 8x$. Moreover $\eta_{\mathrm{rss}} + q^{-4} \le 2u$ and $\eta_{\mathrm{rss}}(1+q^{-2}) \le 2u$.*

Proof. $x \le 1/2$ and $u/x = (2/3)^{m/2} \le 3/8$ hold at $m = 5$ and are monotone in m; $u \le x^2$ is $2^{3/2} \le 3$. $\eta_{\mathrm{rss}} \le \frac{3}{2}u$ is $q^2 - 4q - 1 \ge 0$. $2/(q-1) \le x^2/2$ is $4q^2 \le (q-1)3^m$, true at $m = 5$ ($4096 \le 7533$) and the ratio decreases. $(2+\sqrt{2})^2 q/(q-1) \le (7/2)^2 \cdot 32/31 < 324/25$. Theorem 6.2 gives $\tau^2 \le 5x^2$, so $\nu^2 \le \frac{11}{2}x^2 \le \frac{11}{4}x$ and $\sqrt{2}\nu \le \sqrt{11}\,x \le \frac{10}{3}x$. Splitting the square root in (2) by the triangle inequality, $\theta_2 \le \nu^2 + \sqrt{2}\nu + (\frac{18}{5} + \frac{3}{2})u \le (\frac{11}{4} + \frac{10}{3} + \frac{51}{10}\cdot\frac{3}{8})x = \frac{1919}{240}x$. The last two inequalities follow from $\eta_{\mathrm{rss}} \le \frac{3}{2}u$, $q^{-4} \le u/2$ and $\frac{3}{2}(1+q^{-2}) \le 2$. □

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